Shavaughn Hamilton

Summarized Description of System Design Project:

- Led research and development of a **CRM system**, conducting **risk assessments, security evaluations, and compliance analysis** to ensure data integrity and regulatory adherence.
- Designed and implemented **governance frameworks, security controls, and vulnerability management processes**, emphasizing **risk mitigation and business continuity**.
- Conducted **stakeholder-driven requirement gathering**, feasibility analysis, and **access management planning** to align security strategies with business objectives.
- Developed an **incident response plan, disaster recovery strategy, and total cost of ownership (TCO) assessment**, ensuring long-term **cyber resilience and compliance**.



PHP Agency Customer Relationship Management Tool Creation

Team Member's responsibilities

Team Members	Tasks
Shavaughn Hamilton	Technology Systems Manager
Walker Adlam	Meeting Notes
Sheila Ramos	User Interface Designer
Anika Sultana	Communications & Public Relations

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Phase 1 - Planning Phase

Advanced System Analysis and Design

Letter of Consent

TO WHOM IT MAY CONCERN

This is to certify that Nicole Morales Sancho, liaison of PHP Agency, has permitted and allowed Group 4 in Dr. Pauline Chin's Advanced System Analysis and Design's Fall 2023 class to draft a Customer Relationship Management tool for conducting business.

Liaison for PHP: Nicole Morales Sancho

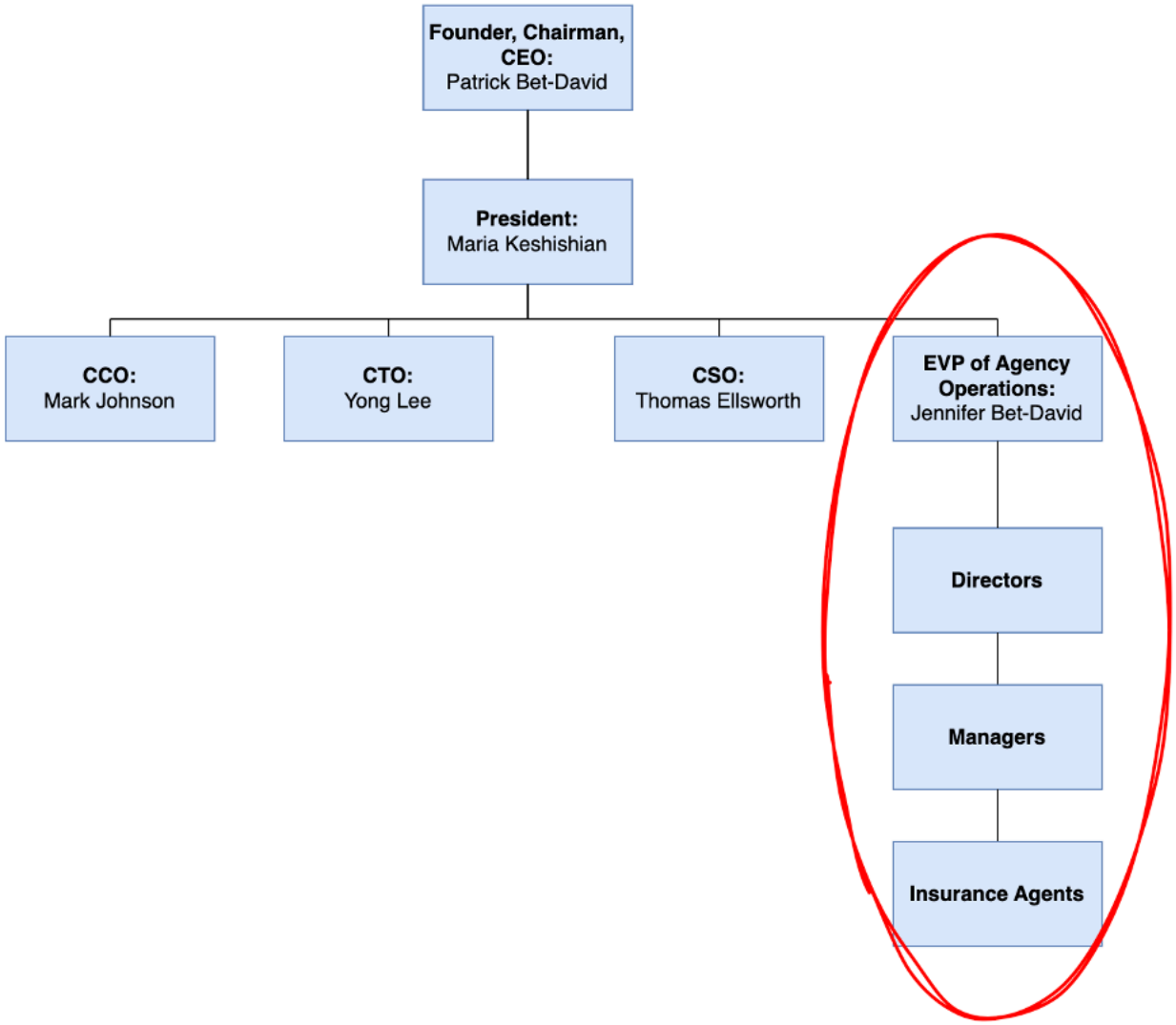
Organization Description

PHP Agency is a national financial services marketing company founded in 2009. With its headquarters situated in Addison, Texas, PHP stands for "People Helping People." The company operates with a mission to "Provide financial education to the underserved communities in America and help protect their families' legacies through life insurance products." PHP Agency prides itself on the diversity of its agents. Their community of agents reflects a broad spectrum of backgrounds, fostering an inclusive environment that prioritizes representation.

The company collaborates with some of the industry's top insurance providers. This enables PHP agents to offer a wide variety of life insurance plans, meeting the diverse needs of their clients.

With over 27,000 agents operating in 49 States and more than 150 offices, PHP has a large enterprise to manage. The company has embraced the use of technology but could benefit from a streamlined system. As it stands, their agents use an array of applications to manage the scope of their work. PHP Agency could improve their workflow through the implementation of a system that integrates all their technology needs in one place, and offers smart training features for new members.

Organizational Structure



***The section circled in red is where our CRM is intended to be relevant in practical use.**

Founder, Chairman, CEO: Patrick Bet-David founded PHP Agency, and serves as the CEO and Chairman. He manages the overall, long-term direction of the company. He is the public face of the company.

President: Maria Keshishian is the President of PHP. She focuses on the day-to-day, daily operations, reports to the CEO, and is second in command. She is focused on internal and private operations.

Chief Commercial Officer (CCO): Mark Johnson is responsible for the organization's commercial development, including activities and growth relating to marketing, sales, product development, and customer service.

Chief Technology Officer (CTO): Yong Lee is the technology executive and leads the technology or engineering department. He develops policies and procedures and use technology to enhance products and services that focus on external customers.

Chief Strategy Officer (CSO): Thomas Ellsworth helps to formulate, facilitate, and communicate the organization's strategic initiatives and future goals.

Executive Vice President of Agency Operations: Jennifer Bet-David is the EVP of Agency Operations and is responsible for coming up with strategies to ensure accomplishment of organizational goals. She appoints departmental heads and assigns tasks to them.

Directors: Manage, coordinate, and supervise business activities of the company. They ensure the company's employees are provided with the best working conditions and that the company has good business relationships with its trading partners.

Managers: Accomplishes department objectives by managing staff; planning and evaluating department activities. Maintains staff by recruiting, selecting, orienting, and training employees. Ensures a safe, secure, and legal work environment. Develops personal growth opportunities.

Insurance Agents: Provides service to clients. Establishes productive working relationships with clients. Develops a base for long-term sources of clients. Compiles lists of prospects.

PHP AGENCY
AN INTEGRITY COMPANY

SWOT Analysis

Strengths

-PHP Agency has a strong company culture.

-PHP Agency has committed members that help recruit and train future agents.

-PHP Agency sells life insurance plans derived from an array of financial service providers. This keeps them competitive and ensures the products they sell are in the client's best interest.

Weaknesses

-Currently limited by the need to manage everything pertaining to new recruits manually.

-Company growth adds strain as more administration positions are needed to preform the tasks of a growing operation.

-The company relies on multiple different applications to run aspects of the operation, and the use of different systems that do not integrate with each other creates inefficiency.

Opportunities

-Improve upon employee training with an automated system that walks new recruits through helpful steps to learn the skills needed.

-Improve upon client management with a CRM, for one source to monitor and automate client engagement.

-Unifying the software PHP Agency uses to improve efficiency, functionality, and ease of use for the agents.

Threats

-The suboptimal efficiency of PHP's current software infrastructure may leave them vulnerable to competitive pressure from companies that employ more advanced and efficient technologies.

-Agents operating under PHP exhibit a high degree of autonomy. Inconsistent practices among these agents could lead to variable performance, potentially impacting client retention.

Mission Statement of PHP Agency:

Fastest growing financial services marketing company in America. We're bringing back life to the insurance industry.

Opportunity Statement

The system to be designed will unify the workflow of PHP Agency, from the agent recruiting process, to the client management scope. The program will enable PHP Agency to grow their operation with confidence, ensuring seamless scalability and organizational agility in the face of expansion, thereby alleviating concerns of operational constraints associated with increased size.

Preliminary Project Plan:

Project Timeline

	Planning	Analysis	Design	Implementation
Standard Distribution	10%	25%	35%	30%
Estimated based on system complexity	Actual:	Estimated:	Estimated:	Estimated:
3 Months total estimate	15%	30%	40%	35%

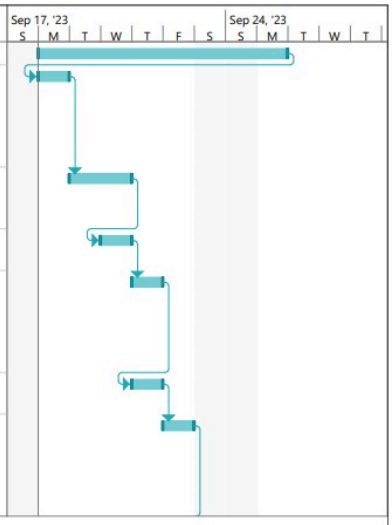
Work Breakdown Structure - Project Work Plan

Work Breakdown Structure						
Task		Phase	Durations		Status	Due
Task #	Task Name	Phase Number	Month	Week	In Progress	Dates
1	Project Planning Phase 1		Sep.	1	Open	9/25
2	Organizing Chart Outlining		Sep.	1	Open	9/25
3	SWOT Diagram		Sep.	1	Open	9/25
4	Mission Statement		Sep.	1	Open	9/25
5	Opportunity Statement		Sep.	1	Open	9/25
6	Preliminary Project Plan		Sep.	1	Open	9/25
6.1	Estimate Project time frame		Sep.	1	Open	9/25
6.2	Identify Project Tasks		Sep.	1	Open	9/25
7	Identify Project Tasks		Sep.	1	Open	9/25
7.1	Determine the Major		Sep.	1	Open	9/25
7.2	Determine Team Member		Sep.	1	Open	9/25
7.3	Create Work Breakdown Structure & Project Work Plan		Sep.	1	Open	9/25
7.4	Create Gantt Charts		Sep.	1	Open	9/25
8	Control and direct project		Sep.	1	Open	9/25
8.1	Develop Project Standards		Sep.	1	Open	9/25
8.2	Risk Assessment Management		Sep.	1	Open	9/25
9	Deliverables		Sep.	1	Open	9/25
10	Project/Team Charter		Sep.	1	Open	9/25
10.1	Team purpose		Sep.	1	Open	9/25
10.2	Scope		Sep.	1	Open	9/25

10.3	Desired End Result		Sep.	1	Open	9/25
10.4	Reporting Plan		Sep.	1	Open	9/25
11	Control and Direct Project		Sep.	1	Open	9/25
11.1	Document Standards		Sep.	1	Open	9/25
11.2	Coding Standards		Sep.	1	Open	9/25
11.3	Procedural Standards		Sep.	1	Open	9/25
11.4	Requirement Standards		Sep.	1	Open	9/25
11.3	Code of Conduct		Sep.	1	Open	9/25
12	Risk Assessment		Sep.	1	Open	9/25
12.1	Risk #1 & Risk #2		Sep.	1	Open	9/25
12.2	Likelihood of risk		Sep.	1	Open	9/25
12.3	Ways to Adress risk		Sep.	1	Open	9/25
13	Preliminary Fact Finding		Sep.	1	Open	9/25
13.1	Potential Interview Questions		Sep.	1	Open	9/25
13.2	Similar Systems		Sep.1	1	Open	9/25
N/A	Requirement Gathering	Phase 2	Oct.	N/A	Not Started	10/9
N/A	Designing the Solution	Phase 3	Oct.	N/A	Not Started	10/30
N/A	Testing & implementing	Phase 4	Nov.	N/A	Not Started	11/13
N/A	System Maintenance	Phase 5	Nov.	N/A	Not Started	11/20
N/A	Final Product	Final Product Phase	Nov.	N/A	Not Started	11/27

Gantt Chart

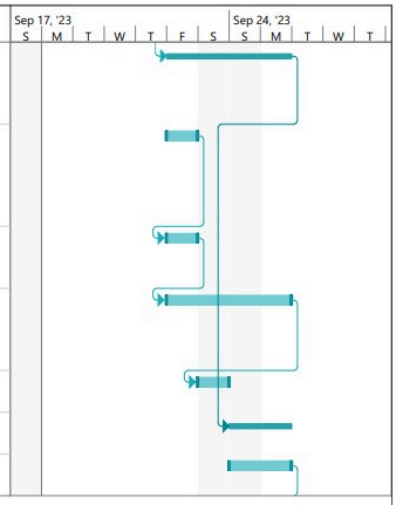
ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		**phase 1**	6 days	Mon 9/18/23	Mon 9/25/23	
2		Organization Chart outlining details of the various functional areas and the reporting	1 day	Mon 9/18/23	Mon 9/18/23	1
3		SWOT Diagram – Analysis for the organization	2 days	Tue 9/19/23	Wed 9/20/23	2
4		Mission Statement of the Organization	1 day?	Wed 9/20/23	Wed 9/20/23	3
5		Opportunity Statement - for the Department this will include a brief statement of the	1 day?	Thu 9/21/23	Thu 9/21/23	4
6		Estimate Project time frame	1 day?	Thu 9/21/23	Thu 9/21/23	5
7		Identify Project tasks, this involves an overview and management of the project	1 day?	Fri 9/22/23	Fri 9/22/23	6



Project: phase 1 gantt chart
Date: Sat 11/25/23

Task	Inactive Summary	External Tasks
Split	Manual Task	External Milestone
Milestone	Duration-only	Deadline
Summary	Manual Summary Rollup	Progress
Project Summary	Manual Summary	Manual Progress
Inactive Task	Start-only	
Inactive Milestone	Finish-only	

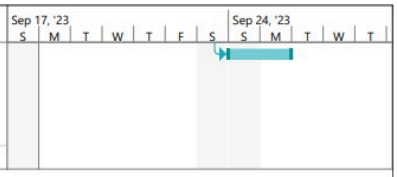
ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
8		**Determine the major components (tasks) that are needed**				**7**
9		Determine team members and responsibilities	1 day	Fri 9/22/23	Fri 9/22/23	
10		Create a Work Breakdown Structure and	1 day	Fri 9/22/23	Fri 9/22/23	9
11		Create Gantt Charts (Organize and sequence these tasks into a	2 days	Fri 9/22/23	Mon 9/25/23	10
12		Control and direct project	1 day	Sat 9/23/23	Sat 9/23/23	11
13		**Develop Project standards**				**8**
14		Risk Assessment and Management	2 days	Sun 9/24/23	Mon 9/25/23	



Project: phase 1 gantt chart
Date: Sat 11/25/23

Task	Inactive Summary	External Tasks	
Split	Manual Task	External Milestone	
Milestone	Duration-only	Deadline	
Summary	Manual Summary Rollup	Progress	
Project Summary	Manual Summary	Manual Progress	
Inactive Task	Start-only		
Inactive Milestone	Finish-only		

ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
15		Deliverables – Diagrams and Narrative details of models developed.	2 days	Sun 9/24/23	Mon 9/25/23	14
16						



Project: phase 1 gantt chart
Date: Sat 11/25/23

Task	Inactive Summary	External Tasks	
Split	Manual Task	External Milestone	
Milestone	Duration-only	Deadline	
Summary	Manual Summary Rollup	Progress	
Project Summary	Manual Summary	Manual Progress	
Inactive Task	Start-only		
Inactive Milestone	Finish-only		

Deliverables: (Details of Solution)
Code and create PHP's own user-friendly application and website that functions as an all-in-one Customer Relationship Management tool. The all-in-one features include:

- Keeping track of interactions, data, and notes about customers or potential
- Text messages
- Threads
- Emails
- Videos

Project/Team Charter

Team Purpose:
The FAU Team, composed of 5 members, will assist and support the office of Nicole Morales Sancho in finding or creating a Customer Relationship Management (CRM) tool/website that will help the company with the recruiting process and managing relationships with customers by keeping track of interactions, data, and notes about customers or potential on a single platform.

Duration and Time Commitment:
The FAU team will work with the office of Ms. Sancho until the end of the 2023 Fall Term. By the end of the semester, the FAU Team would have completed the first three phases of the System Development Life Cycle: Planning, Analysis and Design. Although the FAU Team will not support the office of Ms. Sancho through the Implementation phase and Post Implementation of a Customer Relationship Management Software solution, the FAU Team will provide all the necessary information, so the office of Ms. Sancho has an informed idea of the best solution that fits its office needs.

Scope:
The FAU Team will work with the office of Ms. Sancho to understand the office business needs and based on these, perform the Planning, Analysis and Design phases of the software development phase. At the end of the third phase, the office of Ms. Sancho will have enough information to buy or invest in the creation of the best software solution that meets its needs and budget.

Desired End Results:
Provide Ms. Sancho with the best Customer Relationship Management solution that solves PHP's business needs and adjusts to the office's budget.

Supporting resources:
Ms. Sancho has agreed to stay in contact with the FAU team during each phase should any questions arise and to stay updated on our progress as necessary.

Reporting Plan:
The FAU team has been divided into 5 different roles. Members of the team will perform their assigned tasks and communicate their progress to all team members.

Control and Direct Project (Project Standards)

Procedural Standards:
The FAU Team will conduct at least 15 minutes of daily standup to report progress and discuss issues or concerns. The Project Manager will use a progress report to update the project plan.

Requirement Standards:
All Business Requirement Documents (BRD) will be supplied to vendors. Vendors will use the BRD's to have a general idea of Ms. Sancho's office's needs to prepare before conducting a presentation.

Code of Conduct
All members of the FAU Team shall conduct themselves professionally at all times. No obscene language will be allowed. Not following this Code of Conduct will result in the separation of the members from the group.

Risk Assessment

Risk # 1: A system failure could set the PHP back and lower productivity in the short term due to their need to hire more administrators.

Likelihood of risk: This issue has a low-risk probability, but a high impact if the network were to crash and couldn't be fixed quickly.

Ways to address risk: Beyond the need to hire more administrators which could solve the problem, the addition of a help desk feature that offers technical support to current employees would be useful to ensure that the problems can be fixed.

Risk # 2: A system of this size could go over budget and set PHP agency back.

Likelihood of risk: This issue has a medium risk probability, but a high impact if the agency were to go far enough over budget.

Ways to address risk: Implementing and maintaining a company-wide information system could prove to be a substantial cost for the agency. We can use other businesses that are similar

in size, that have developed their own system as a reference for how much it will cost to create and maintain. Conducting proper project monitoring and project reporting techniques will also ensure that managers are aware of the progress and the cost of the new system as it is being worked on.

Preliminary Fact Finding

Potential Interview Questions:

- What would PHP Agency's budget be for implementing a new company-wide CRM system?
- Which specific tools and functions would this application be replacing?
- How many customers would need to use the platform on a weekly basis?
- If the system were to crash, would the company have a response plan in place?

Similar systems:

- Salesforce
- HubSpot

Phase 2 - Requirement Gathering

Advanced System Analysis and Design

Detailed Fact-Finding

Stakeholders Pertaining to Proposed System

The system in question will impact the complete array of stakeholders for PHP Agency. The system will maximize efficiency and add functionality to the normal operations within the scope of PHP employee demands, but it will also add functionality and clarity to the customer experience at PHP. Two user interface designs will be implemented, one for current and potential employees at PHP, and one for customers to streamline their process.

Insurance Agents 	**Overview:** Insurance agents use information that they gather from multiple sources to get a summary of the status of customers and potential customers. It will also aid in future decision making geared towards improving employee turnover. **Stakeholder Type:** Internal – Assists with internal customer issues. **Interaction:** Interacts directly with potential customers
Managers 	**Overview:** Managers use information from the system to track interactions between insurance agents and acquired customers, and analyze performance related to new acquisitions and retention. **Stakeholder Type:** Internal – Regularly interacts with the system to oversee insurance agent and customer relations at the facility. **Interaction:** Secondary authority over insurance agents, with interaction at all levels, from between insurance agents, and answers to Director

Directors 	**Overview:** An experienced, senior-level manager who is responsible for the overall success of the office they have been assigned to by executive positions. **Stakeholder Type:** Internal – Regularly interacts with the system and oversees all internal operations at the facility. **Interaction:** Total authority within the branch they oversee

Fact Finding Process

A thorough analysis of system requirements has been conducted. Discussions have been held with PHP employees to attain an idea of their current workflow strategies, and areas of potential improvement. PHP Agency currently uses a simple CRM system they designed, named "Bamboo." This system is extremely limited in functionality, only serving as a database for information about customers. The PHP agents envision a broad system that can take on functionality that improves many aspects of the organizational flow.

First, we identified in detail the current operational process, from the new employee recruitment process to the client management side of demands. Then, we walked through the steps with the employees and suggested proposed solutions to ensure we were corroborating their vision for the new system. After analyzing the current system, and the suggested pitfalls and inefficiencies with the PHP system, we have been able to design a clear set of requirements that will meet PHP Agency's needs entirely.

II) Interview Questions

Interview Questions
1) What software does your company currently use? "Bamboo" to automate some operational tasks and paperwork. "Suran eBay" for onboarding and compliance. Google apps like Google docs, Google sheets, and Gmail for file sharing. Zoom and WhatsApp for live communication and meetings. "Monday.com" Social media platforms like Instagram, Facebook, and Twitter to market and communicate. Wind flex web for illustrations for certain carriers **2) What's something you wish existed to make things easier for your company, but may not exist yet?** A tool that would help with the recruiting process like some sort of AI that guides customers through the process in steps saying so that it takes off some pressure from the administrative work in the recruiting process. Also, a tool that connects agents with the right people, has group text capabilities, and displays phone numbers. A tool that gives quick accessibility, shows employees and customers their platform systems, and gets them integrated with everything that comes through PHP. From the clientele side, implementing a Customer Relationship Management tool would help. **3) What problems do you want this new system to solve?** Optimizing the administrative work because their onboarding process has gotten a lot bigger, speeding up that has slowed them down. At times we don't have the capacity to maximize our recruiting because there's a lot of little things that go in between and not every single new person knows how to transfer that to the other person, which makes communication fall through the cracks and leaves certain clients not being able to get certain things. Streamlining the onboarding process, to get everyone going through filters quicker. (that's where they don't have software like that yet) they must utilize a lot of different things.

III) Survey Questionnaire

1. What is your preferred way of communicating with clients?

Text Messaging
Video
In-Person
Email

2. What is your preferred way of communicating with co-workers?

Text Messaging
Video
In-Person
Email

3. On average, how many clients do you communicate with daily?

1-4
5-8
9-12
Over 12

4. On average, how many co-workers do you communicate with daily?

1-4
5-8
9-12
Over 12

5. Is it harder to acquire clients or to retain clients?

Acquire clients.
Retain clients.

6. Rate the efficiency of your current technological workflow.

Bad
Fair
Good
Excellent

	A1	A2	A3	A4
Q1	4	2	12	2
Q2	3	1	15	1
Q3	3	8	7	2
Q4	3	12	2	3
Q5	20	0	0	0
Q6	4	12	4	0

Based on 20 employees' feedback

IV) Detailed Observation of Existing Work Processes

This PHP Agency office opens from 9 AM – 9 PM daily from Monday to Saturday and is closed on Sundays. The work process at PHP includes training 2 days a week. It is a commission-based job, and the amount of involvement in various aspects is wholly dependent on the individual's needs rather than a requirement. Team mentorship is an important aspect of learning from the success and mistakes of others in the company with more experience. There are days when new employees can shadow an experienced one in meetings to learn successful habits.

Individual agents who gain a handle and are confident in their skills can work from home independently. In the office, there are team managers available to contact regarding questions. They assist with the administrative work after new clients are acquired. In the office, agents can practice following leads and landing new clients while being advised by their team members. Employees use various software applications such as Bamboo to stay organized and be successful in each task. There are various conventions that team members can attend to learn from individuals that are successful within the PHP agency.

The first stage is gathering leads who are interested in using PHP's services. The next stage is setting up an appointment to meet with potential clientele. During the appointment, which is either virtual, a public place, or the potential client's home, the agents educate the prospective clientele on PHP's services. After potential clientele is informed, they organize a personal plan for them if they are interested, they send files, documents, and contracts through email if the meeting was over the phone or virtual for them to sign up. If the processes started in person, some contracts can be done in person if that was the mode of the meeting. If the potential clientele agrees to do business with PHP, an agent will follow up with completing the process in the upcoming days if necessary.

Use Case Descriptions

A dual-path system will be introduced, comprising the "Employee Portal" and the "Client Portal." Each path is designed with distinct functionalities and objectives.

Employee Portal Overview:

PHP Agency is keen on streamlining the employee recruitment and task management processes. The current recruitment procedure lacks organization and requires synergy from many existing employees. Often the method of organizing events involves texting or emailing employees to

determine who can attend. New recruits are left to decipher the process without a centralized guiding system.

PHP Agency emphasizes a need for simplicity in the system functionality, desiring a user-friendly design.

Employee Portal

Login
Employees and prospective recruits will access their dashboard using a unique Employee ID and Password.

The Employee dashboard will display the main information for the tab currently selected from the available options.

The tabs included on the Employee Dashboard:
- My Profile
- My Clients
- Calendar
- Training

The page will include a box for notifications that can be minimized.

My Profile
The "My Profile" page will display the Employee user information that is held in the database for employees. This page will not be frequently used by employees but is there to allow employees to interact with their profile and change basic information like name and login credentials.

My Clients
The "My Clients" page will help PHP Agents manage all their customer interaction needs. The page will first display a list of clients with an assigned status, "Active." All clients within an employee database will have an assigned status. 'Past', 'Potential', and 'Active' status changes can be made at any time.

Organizing the My Clients page by status of client will enable employees to quickly complete the necessary task for the given clients. 'Active' clients are ones currently taking steps to purchase an insurance plan with a PHP Agent. This process takes time, and requires document uploads, reviews, and "illustrations" of insurance plans to choose from based on unique factors for each client.

Active
In order for a client to be in 'Active' status, they must have a corresponding Client login and Client dashboard. The employee "My Clients" page allows agents to click on 'Active' clients to interact with their customer's dashboard. Employees can send notifications to clients to request

information upload, display insurance options, and add updates on the process of their insurance purchase.

Potential and Past

Found under the ‘Active’ status clients, all clients that are in an agent's database that are not ‘Active’ are either in ‘Past’ or 'Potential’ Status. This section will help employees keep a record of the deals they have worked on and gives automation functionality. Here, we have a “Draft Outreach” option that lets the employee build either a text or email message. After building the outreach message, the employee can select which clients should receive the mass message.

Calendar

The “Calendar” page is crucial to employee operations management, and new recruit needs. The page will first load with the total calendar for the PHP Agency local network, but will enable filters, so the employees can see their own events and obligations. The goal of the Calendar functionality is to produce a unified space to organize employees for events and new recruit training. Here, employees can create new events, and even invite other members to their events. This allows members in managerial positions to send out mass event invitations, and individual employees to build events and obtain confirmation from those who are available to join.

Training

The training page will be useful for both new recruits and seasoned PHP Agents. The initial functionality will enable new recruits to navigate the steps needed to become a PHP Agent. From this page, they can request help from local agents. If a new recruit requests help from local agents, an event is sent out as a notification to other PHP Employees with an attached calendar invitation, here employees can accept or decline the invitation to meet and help the new recruit.

Without this system, all of these needs are currently organized without order or consistency. This system will enable the most efficient organization of new employee training, and it will help keep a record of agents who are taking on extra responsibilities to help new recruits.

The training page will include basic information to teach new PHP Agents the skills they need, and the upper management can upload new documents and videos that are relevant to all PHP Agents.

Client Portal

Login

Clients will log into their Client Portal with a unique Client ID and Password.

The Client Portal will be simple and offer clients of PHP Agency a one-stop shop for purchasing insurance. The Client Portal will always open up to the “Home” page and will have the option to navigate to their “My Profile” page.

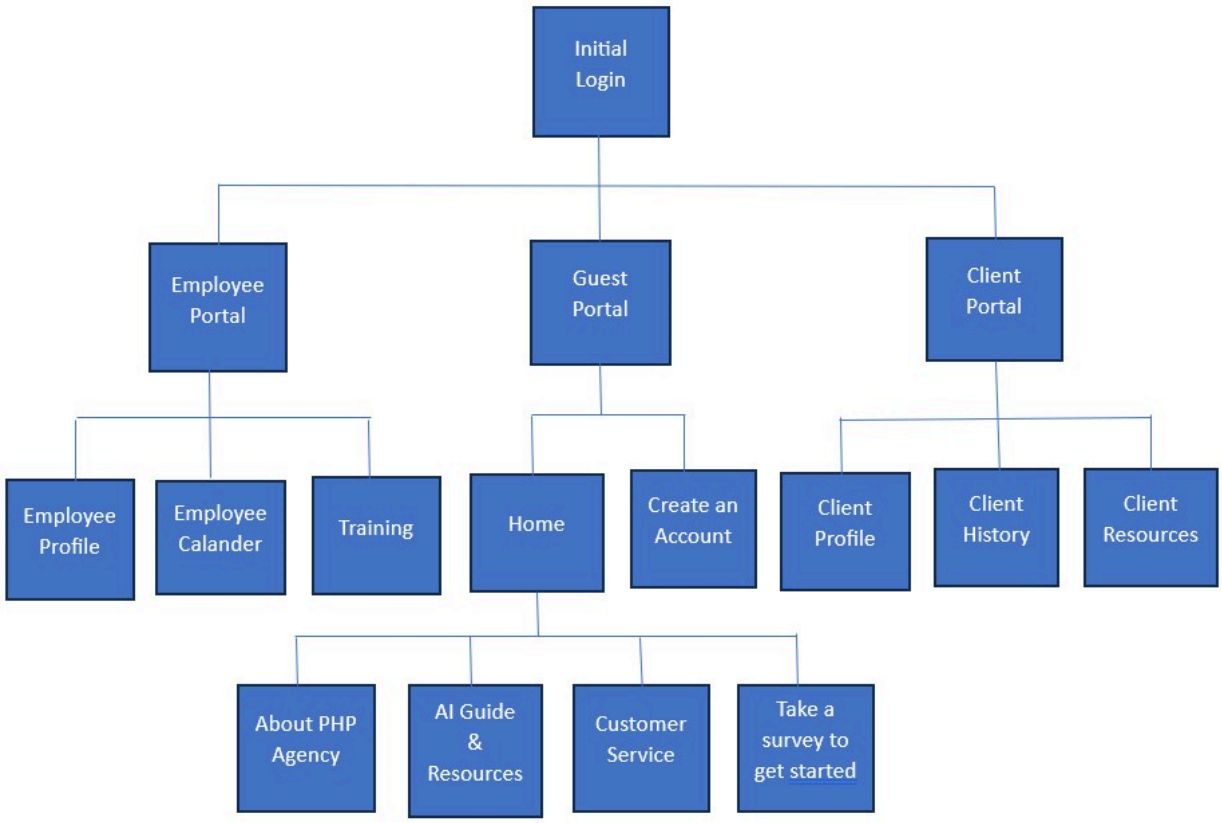
My Profile

The “My Profile” page in the client portal allows clients to view and change their contact information, name, and family information that is relevant to the insurance policies.

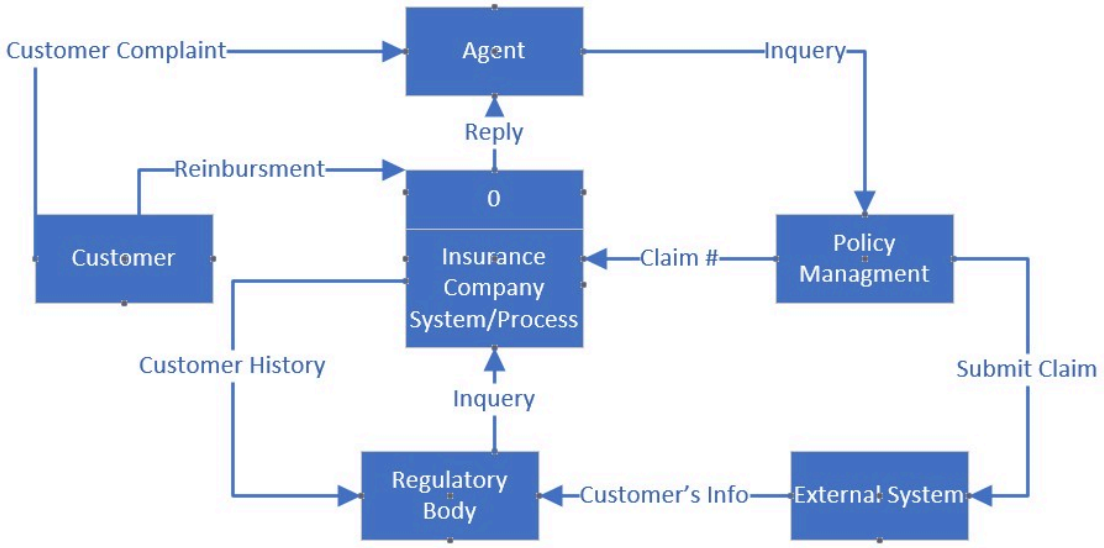
Home

The “Home” page in the client portal allows the client to engage with all the needed steps of purchasing insurance. It will display a status for their process, and it will allow them to review insurance plans. Here, clients can upload documents and relevant information when they are prompted to do so. Once they have completed the purchase of an insurance plan, the client can log into the portal to review their plan in depth, and request help from an agent if they have any questions.

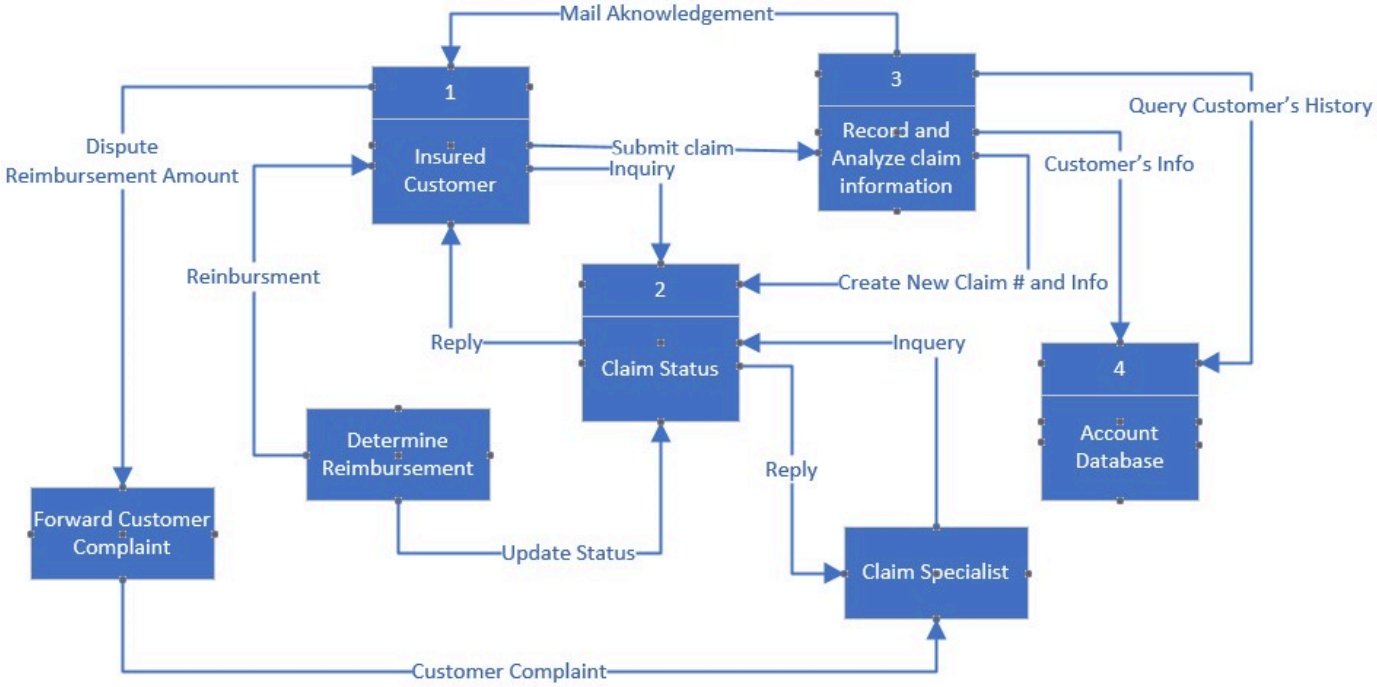
Functional Decomposition Diagram



Customer/Employee Context Diagram



Customer/Employee Data Flow Diagram



Feasibility Analysis

Market Analysis:

Target Market:

- Insurance apps can cater to a wide range of users, including policyholders, agents, and employees.
- Analyze market trends to identify potential opportunities, such as a growing demand for digital insurance services.

Competition:

- Conduct a competitive analysis to understand the strengths and weaknesses of existing insurance apps.
- Identify gaps in the market, such as specific insurance types or features not adequately covered by competitors.

Regulatory and Compliance:

- The insurance industry is heavily regulated to protect consumers. Ensure the app complies with industry-specific regulations and data protection laws.
- Establish strict data privacy measures to safeguard customer information.

Technical Feasibility:

Technology Stack:

- Choose the technology stack based on the platform(s) you intend to target (e.g., native mobile apps for iOS and Android, web-based).
- Consider factors like scalability, security, and ease of maintenance when selecting technologies.

Data Security:

- Implement encryption protocols to protect customer data during transmission and storage.
- Regularly conduct security audits and penetration testing to identify and fix vulnerabilities.

Integration:

- Ensure that the app can seamlessly integrate with the insurance company's existing systems, databases, and third-party services.
- Develop APIs or use industry-standard protocols for data exchange.

Financial Feasibility:

Budget and Resources:

- Estimate development costs accurately, including salaries, software licenses, hardware, and infrastructure.
- Consider ongoing operational expenses for maintenance, server hosting, and updates.

Revenue Model:

- Determine the most suitable revenue model for the app, such as subscription-based, freemium, or advertising-supported.
- Perform financial modeling to project revenue based on user growth and pricing strategies.

Return on Investment (ROI):

- Calculate ROI by comparing expected revenue with initial and ongoing costs.
- Set clear milestones to measure the app's performance and adjust strategies as needed.

Operational Feasibility:

Workflow Analysis:

- Collaborate closely with the insurance company to map out existing processes.
- Identify areas where the app can streamline tasks, such as claims processing or policy management.

Employee Training:

- Ensure that employees are trained to use the app effectively and provide ongoing support.
- Develop training materials and user guides to facilitate adoption.

Legal and Ethical Considerations:

Privacy and Data Protection:

- Define clear data handling policies and practices to comply with data protection laws (e.g., GDPR, HIPAA).
- Obtain informed consent from customers regarding data collection and usage.

Intellectual Property:

- Protect the app's intellectual property through patents, trademarks, or copyrights.
- Ensure that there are no infringement issues with existing patents or trademarks.

Risk Analysis:

Technical Risks:

- Identify potential technical challenges, such as integration complexities or software bugs, and create risk mitigation plans.
- Establish regular testing and quality assurance procedures.

Market Risks:

- Monitor the market for changing trends, competition, and shifts in customer preferences.
- Be prepared to pivot or adapt the app's features based on market dynamics.

Security Risks:

- Recognize that data breaches or security vulnerabilities can have severe consequences. Invest in robust security measures, conduct regular security audits, and have a well-defined incident response plan.

Marketing and Promotion:

Marketing Strategy:

- Develop a comprehensive marketing plan that includes digital marketing, content creation, and social media campaigns.
- Consider partnerships with influencers or industry organizations for promotion.

User Acquisition:

- Identify strategies for acquiring users, such as referral programs, advertising, or incentivized sign-ups.
- Focus on user retention through regular engagement and updates.

User Experience and Design:

User-Centered Design:

- Prioritize user experience (UX) by conducting user research and usability testing.
- Create a user interface (UI) that is intuitive and aesthetically pleasing.

Usability Testing:

- Continuously gather feedback from users through beta testing and post-launch reviews.
- Use this feedback to iterate and improve the app's design and functionality.

Sustainability:

Long-term Support:

- Ensure that there is a plan in place for maintaining and updating the app to keep it relevant and secure.
- Allocate resources for ongoing customer support.

Scalability:

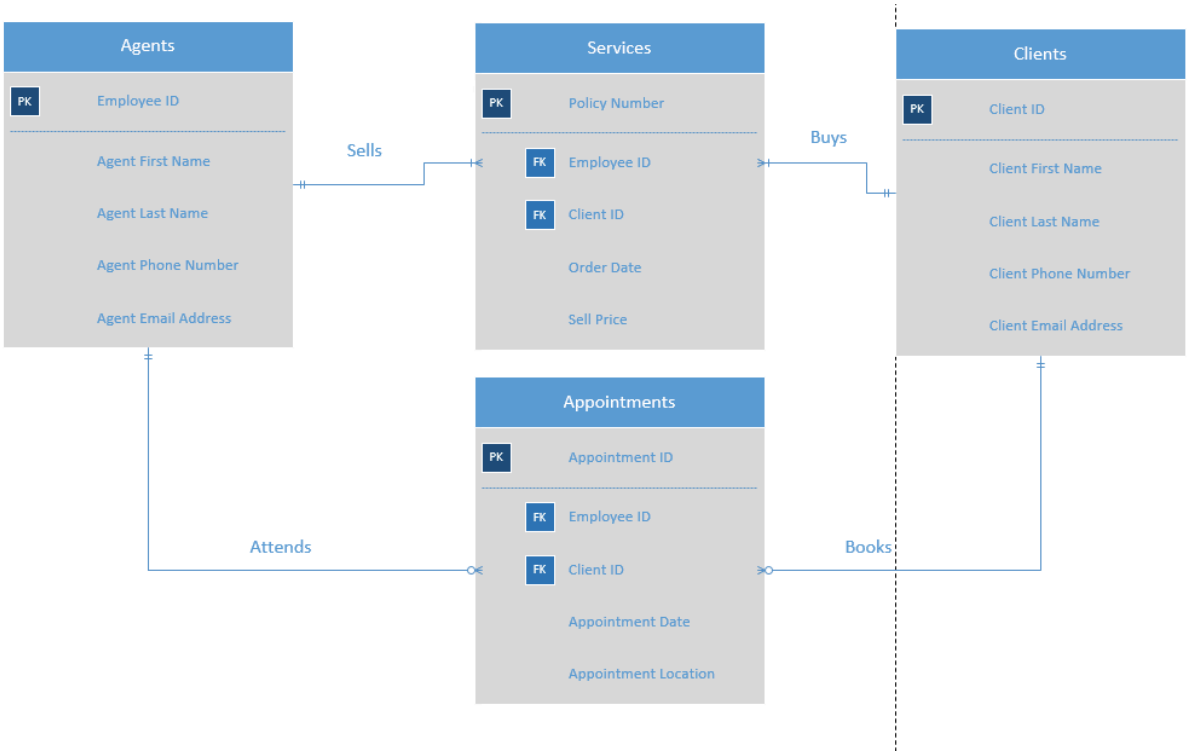
- Design the app architecture to accommodate potential increases in user volume and additional features.
- Monitor performance and scalability as the user base grows.

Environmental Impact:

Green IT Practices:

- Energy-efficient hosting options, minimize server usage, and optimize code for sustainability.
- This software will increase paperless practices in insurance processes to reduce environmental impact.

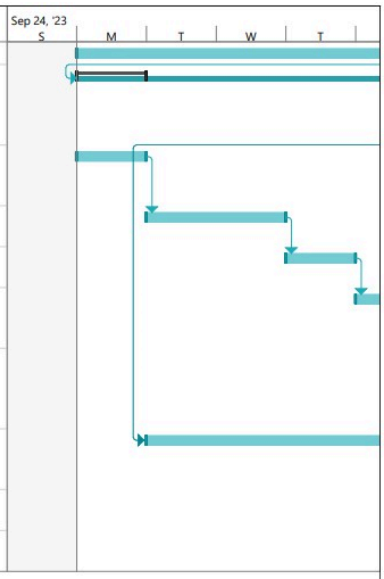
Entity Relationship Diagram



The entity relationship diagram shows how our system will be helpful to employees of PHP Agency, as well as customers and upper management. This diagram shows the entities that will need data stored within our system and how they interact with each other. Agents sell life insurance services to clients who purchase them for the agency. Both clients and agents of the company will be able to access the system to view their upcoming appointments.

Gantt Chart Project Plan

ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		phase 2	13 days	Mon 9/25/23	Wed 10/11/2	
2		**Do detailed fact-finding to determine the requirements.**	**1 day?**	**Mon 9/25/23**	**Mon 9/25/23**	**1**
3		Verify and fully identify all key stakeholders	1 day	Mon 9/25/23	Mon 9/25/23	
4		Further develop interview	2 days	Tue 9/26/23	Wed 9/27/23	3
5		Schedule and conduct	1 day	Thu 9/28/23	Thu 9/28/23	4
6		Develop and distribute surveys/questionn	1 day	Fri 9/29/23	Fri 9/29/23	5
7		Detailed observation of existing work processes	3 days	Mon 10/2/23	Wed 10/4/23	6
8		Fully Developed Feasibility Analysis	4 days	Tue 9/26/23	Fri 9/29/23	2
9		Fully Developed Use Case Descriptions	1 day	Mon 10/2/23	Mon 10/2/23	8
10		Develop Data Flow Diagrams	1 day	Tue 10/3/23	Tue 10/3/23	9



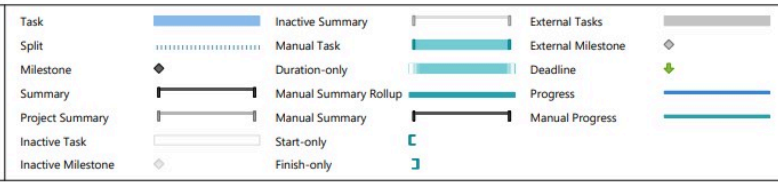
Project: Project3
Date: Sat 11/25/23

Task
Split
Milestone
Summary
Project Summary
Inactive Task
Inactive Milestone
Inactive Summary
Manual Task
Duration-only
Manual Summary Rollup
Manual Summary
Start-only
Finish-only
External Tasks
External Milestone
Deadline
Progress
Manual Progress

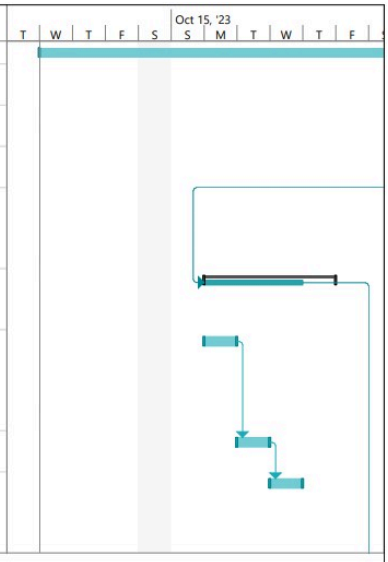
ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
11		Develop Entity Relationship Diagrams	1 day	Wed 10/4/23	Wed 10/4/23	10
12		Detailed Microsoft Project Management Gantt Chart Project Plan	4 days	Thu 10/5/23	Tue 10/10/23	11
13		Deliverables – Diagrams and Narrative details of models developed.	2 days	Wed 10/11/23	Thu 10/12/23	12



Project: Project3
Date: Sat 11/25/23



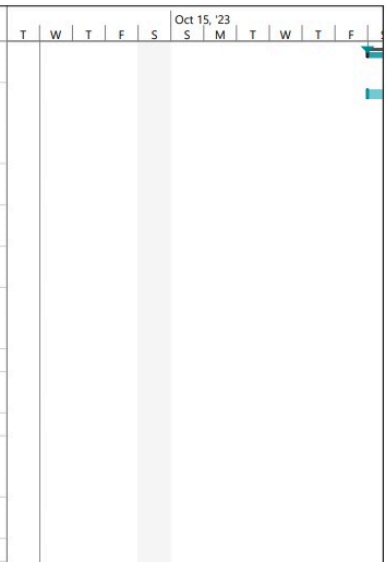
ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		phase 3	20 days	Wed 10/11/2	Fri 11/3/23	
2		**System Acquisition Strategy**	**1 day?**	**Thu 11/2/23**	**Fri 11/3/23**	**1**
3		Create an Alternative matrix				
4		**Designing the environment.**	**2 days?**	**Sat 10/28/23**	**Tue 10/31/23**	**2**
5		This involves determining and identifying all the technology				
6		**Designing the Application Architecture**	**4 days?**	**Sun 10/15/23**	**Thu 10/19/23**	**4**
7		Application structure, partitioning the application in to sub-	1 day?	Sun 10/15/23	Mon 10/16/23	
8		Database infrastructure	1 day?	Tue 10/17/23	Tue 10/17/23	7
9		Access controls, remote wireless access via internet.	1 day?	Wed 10/18/23	Wed 10/18/23	8



Project: Project4
Date: Sat 11/25/23

Task
Split
Milestone
Summary
Project Summary
Inactive Task
Inactive Milestone
Inactive Summary
Manual Task
Duration-only
Manual Summary Rollup
Manual Summary
Start-only
Finish-only
External Tasks
External Milestone
Deadline
Progress
Manual Progress

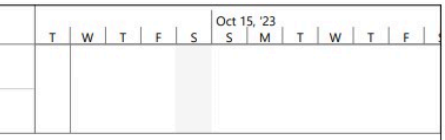
ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
10		**Designing the User Interface.**	**4 days?**	**Fri 10/20/23**	**Thu 10/26/23**	**6**
11		Graphical User Interface Design, Website design, hand-held	1 day?	Fri 10/20/23	Mon 10/23/23	
12		Dialog and Storyboard design	1 day?	Tue 10/24/23	Tue 10/24/23	11
13		Input, output, report designs	1 day?	Wed 10/25/23	Wed 10/25/23	12
14		**Software/Program Design**	**4 days?**	**Mon 10/30/23**	**Thu 11/2/23**	**10**
15		Application Software logical design	1 day?	Mon 10/30/23	Mon 10/30/23	
16		Structural Chart	1 day?	Tue 10/31/2	Tue 10/31/2	15
17		Program Specification	1 day?	Wed 11/1/23	Wed 11/1/23	16
18		Pseducode	1 day?	Thu 11/2/23	Thu 11/2/23	17
19		**Designing the database controls and security**	**3 days?**	**Sun 10/29/23**	**Tue 10/31/23**	**14**
20		Designing Integrity Controls	1 day?	Sun 10/29/23	Sun 10/29/23	
21		Input Controls	1 day?	Sun 10/29/2	Sun 10/29/2	20
22		Output Controls	1 day?	Sun 10/29/2	Sun 10/29/2	21



Project: Project4
Date: Sat 11/25/23

Task
Split
Milestone
Summary
Project Summary
Inactive Task
Inactive Milestone
Inactive Summary
Manual Task
Duration-only
Manual Summary Rollup
Manual Summary
Start-only
Finish-only
External Tasks
External Milestone
Deadline
Progress
Manual Progress

ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
23		Designing Security Controls	1 day?	Thu 11/2/23	Thu 11/2/23	22
24		Disaster and Recovery Plan	1 day?	Thu 11/2/23	Thu 11/2/23	23



Project: Project4
Date: Sat 11/25/23

Task	Inactive Summary	External Tasks
Split	Manual Task	External Milestone
Milestone	Duration-only	Deadline
Summary	Manual Summary Rollup	Progress
Project Summary	Manual Summary	Manual Progress
Inactive Task	Start-only	
Inactive Milestone	Finish-only	

Phase 3 - System Design

Advanced System Analysis and Design

“PHP Agency”

System Acquisition Strategy

Alternative Matrix
In the figure below, it shows the alternative matrix used to compare and contrast the different solutions that might be implemented into the new PHP Agency system. The system that results in the highest score will be the best one for PHP to use. After analyzing the matrix, it can be seen that the new system appears to be the best solution for PHP Agency.

	PHP Agency’s Legacy System		**PHP Agency’s New System**	
Criterion	**Weight**	**Score**	**Weight**	**Score**
Cost	3	2	3	2
Efficiency	4	3	10	9
Development Time	5	5	7	6
Integration	2	2	8	7
Total Score	**14**	**12**	**28**	**24**

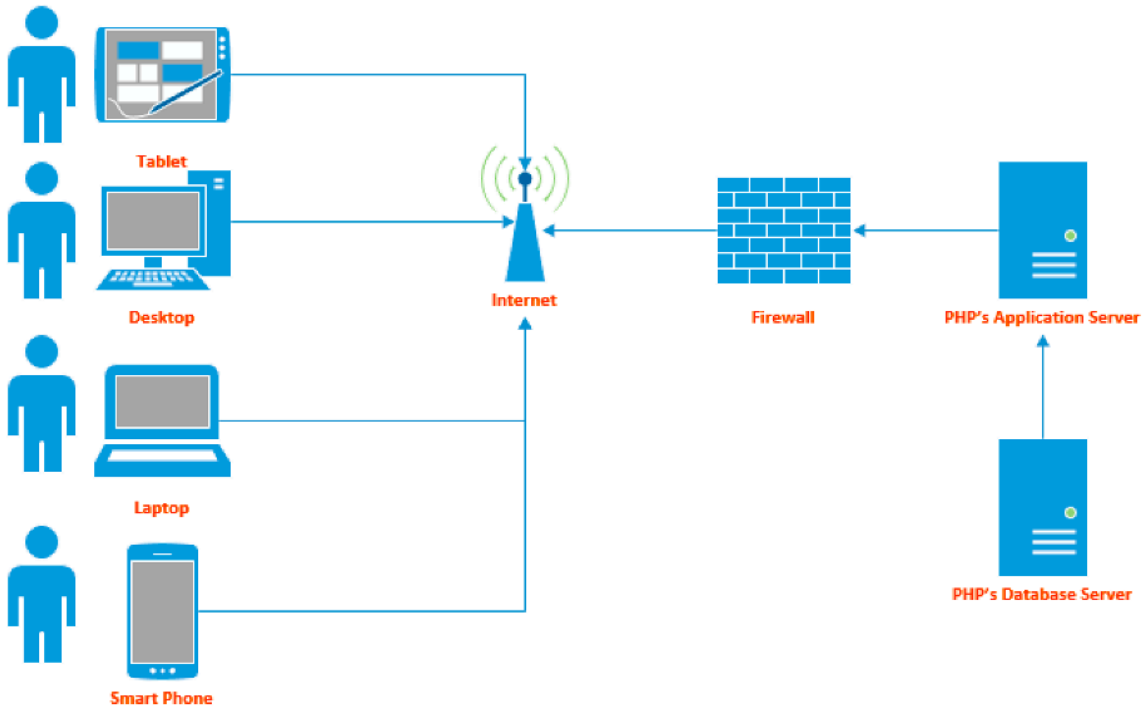
Environment Design

PHP Agency is heavily reliant on technology used to conduct daily business. Directors and managers have access to the database of all clientele so that their information, documents, agreements, communications, and contracts can be processed and archived for future reference. These files need to be saved with redundancy to mitigate loss of information, so information added needs to be automatically updated and backed up at least every hour on various computers and hard drives, both on and offsite. This means strong wireless internet connection and security is a necessity. Since PHP’S organization and communication would be primarily conducted through our system, computers and laptops should be available for any agent in the office, and a company laptop, cellular device and tablet should be available for employees when they work remotely. All these devices would be for business purposes, and the information and data they gain will be tracked by the company in an effort to help with the acquisition and retention of customers. With these devices sharing information and communication plans with each other, the success of all employees will be increased exponentially in terms of efficiency and organization. PHP currently uses Windows products, but the website and application would be compatible With Apple and Linux if an employee so chooses to use their own devices to login to the system and no information would be lost if the company were to change from the current Windows operating system. All the routers, firewalls, computer servers, and desktops are already in place,

so the only things that would need to be added are the cellular devices and laptops for working remotely while staying productively connected.

Application Architecture

Application Structure



We need to set the environment to begin building. For easier understanding, we have created a table with the environment needed and a brief description of it.

Environment Description

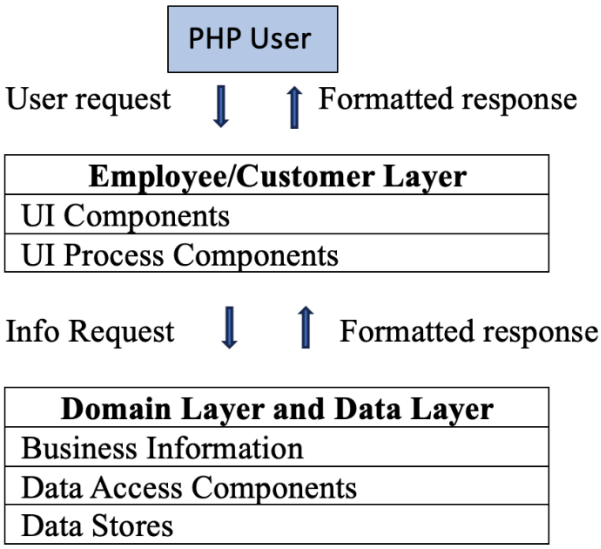
Environment	Type of Environment Required	Description
Database Server	Technical, System Integration, Maintainability	Computer that provides database services to other computers
Application Server	Technical, System Integration, Maintainability	Computer that hosts the software that delivers our application through communication protocol.

Desktop Computers	System Integration, Maintainability	Any desktop computer that might be used by a staff member
Tablets	System Integration, Portability, Maintainability	Any tablet available with iOS, Android or Windows and internet connectivity
Smartphones	System Integration, Portability, Maintainability	Any smartphone available with iOS, Android or Windows and internet connectivity
Firewalls	Technical, System Integration, Maintainability	Security designed to block unauthorized access while permitting outward communication
Routers	Technical, System Integration, Maintainability	Disperses the signal to devices on the network
Communication Protocols and Systems	Technical, System Integration, Maintainability	HTML, TCP/IP, TLS
Operating Systems	Technical, System Integration, Portability, Maintainability	Windows 7, Windows 8, iOS, Android, Chrome OS
Web browsers	Technical, System Integration, Portability, Maintainability	Internet Explorer, Mozilla Firefox, Google Chrome, etc.
Frameset	Technical, System Integration, Portability, Maintainability	JavaScript, Active X Control

Designing the Application Architecture and Software

This is a web-based system that should be divided into 2 layers: Employee Portal/Customer Portal and the Domain/Data Layer

- The employee/customer layer accepts user input and formats and displays processing results for information useful to the PHP employee, including Help, Email, Notifications, Profile, Service, Current Clients, Schedule, About PHP Agency, Training, Company News, and Resources.
- The domain/data Layer implements business information. The data layer manages the stored data for organization and communication of the business such as employee name, address, contact information, etc.



The system should contain a total of nine subsystems for each (employee or customer). There should be a subsystem for

Employee:
Help with FAQ, Help Email, Hotline Number
Profile with Profile Picture, First/Last Name, Phone Number, Employee Number
Training with Company Assigned, Training
Email with Compose, Inbox, Sent, Deleted
Current Clients with List of Current Clients
Company News with List of Recent, Company-wide, News and Events
Notifications with Recently Received Emails, Recently Added Training, Recently Added Appointments, Recently Added Company News
Schedule with Calendar
Resources with Files, Phone Numbers, Email Addresses

Customer:
Help with FAQ, Help Email, Hotline Number
Profile with Profile Picture, First/Last Name, Phone Number, Customer Number
About PHP with Mission Statement, Facts, and Information about the company
Email with Compose, Inbox, Sent, Deleted
Services with List of All Services provided by PHP
Company News with List of Recent, Company-wide, News and Events
Notifications with Recently Received Emails, Recently Added Training, Recently Added Appointments, Recently Added Company News
Schedule with Calendar
Resources with Files, Phone Numbers, Email Addresses

Application Software Logical Design

1) Create a new Account

1. Employee/User – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Create New Account.”
4. System – Displays “Create New Account” page
5. Employee/User - User enters all the following information that is prompted:
6. Create a New Account
    - First Name
    - Last Name
    - Email
    - Username
    - Password
    - Address
7. Login
    - Username
    - Password
    - Forgot Username/Password
8. System – Accepts User information and generates an email to user for password set-up
9. Employee/User - receives an email address with link
10. Employee/User - Clicks link which brings up a password set-up Web page
11. Employee/User - Enters a password
12. Employee/User - Enters a password again to confirm

2) Create Profile

1. Employee/Customer – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Log In.”
4. System – Displays “Log-In” page
5. Employee/User – Enters Log-In ID and Password
6. System – Displays a staff member home page
7. Employee/User – Selects Create Profile
8. Employee/User – Selects “Profile Picture.”
9. Employee/User – Uploads Picture from Device
10. Employee/User – Enters First and Last Name
11. Employee/User – Selects Confirm if everything looks correct

3) Training (Employee only)

1. Employee/Customer – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Log In”

4. System – Displays “Log-In” page
5. Employee/User – Enters Log-In ID and Password
6. System – Displays a staff member home page
7. Employee/User – Selects Company Assigned Training
8. System – Displays Training Modules/Videos
9. Employee/User selects Module of choice

4) Email

1. Employee/Customer – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Log In”
4. System – Displays “Log-In” page
5. Employee/User – Enters Log-In ID and Password
6. System – Displays a staff member home page
7. Employee/User – Selects Email
8.System- Displays Compose, Inbox, Sent, Deleted
9. Employee/user selects displayed links to complete desired action

5) Current Clients (Employee only)

1. Employee/Customer – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Log In”
4. System – Displays “Log-In” page
5. Employee/User – Enters Log-In ID and Password
6. System – Displays a staff member home page
7. Employee/User – Selects List Current Clients
8. System – Displays List of Current Clients with Communication Threads and Information

6) Company News

1. Employee/Customer – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Log In”
4. System – Displays “Log-In” page
5. Employee/User – Enters Log-In ID and Password
6. System – Displays a staff member home page
7. Employee/User – Selects Company News
8. System Displays- Thread of Company News

7) Notifications

1. Employee/Customer – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Log In”
4. System – Displays “Log-In” page
5. Employee/User – Enters Log-In ID and Password
6. System – Displays a staff member home page
7. Employee/User – Selects Notifications
8. System- Displays Notifications

8) Schedule

1. Employee/Customer – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Log In”
4. System – Displays “Log-In” page
5. Employee/User – Enters Log-In ID and Password
6. System – Displays a staff member home page
7. Employee/User – Selects Schedule
8. System – Displays Schedule Calendar
9. Employee/User – Enters Dates of Appointments
10. Employee/User – Enters Continue Button
11. Employee/User – Enters Continue button
12. Employee/User- Looks over Review Page and selects “Edit” if there is a change needed or select “Confirm” if everything looks good
13. Employee/User – If “Edit” user is taken back to “Enter Schedule Availability” and can make any necessary changes to dates and shifts available to work
14. Employee/User – If “Confirm” a note will show up those states “an email confirmation will be sent.”

9) Resources

1. Employee/Customer – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Log In”
4. System – Displays “Log-In” page
5. Employee/User – Enters Log-In ID and Password
6. System – Displays a staff member home page
7. Employee/User – Selects Resources
8. System- Displays Files, Phone Numbers, Email Addresses
9. Employee/User selects desired resource

10) About PHP Agency (Customer only)

1. Employee/Customer – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Log In”
4. System – Displays “Log-In” page
5. Employee/User – Enters Log-In ID and Password
6. System – Displays a staff member home page
7. Employee/User – Selects About PHP Agency
8. System- Displays Mission Statement Facts and Information about the company

11) Services (Customer only)

1. Employee/Customer – From a Web browser enters the web address for the scheduling system
2. System – Displays home page of website
3. Employee/User – Clicks on “Log In.”
4. System – Displays “Log-In” page
5. Employee/User – Enters Log-In ID and Password
6. System – Displays a staff member home page
7. Employee/User – Selects Services
8. System- Displays List of Services provided by PHP Agency
9. Employee/user selects desired service

User Interface Design

User Interface Design Overview

Layout: The system should have two layouts, one for employees, and one for customers. The employee portal should allow access to an employee’s profile, current client, their schedule, and a training tab as well as a notifications tab at the top of the page. The client portal will also include a profile section and a calendar for scheduled appointments as well as information about PHP Agency and what they offer.

Content Awareness: Users should be able to clearly see the content that is available to them and be able to quickly navigate the system to find the information that they intend to find.

Aesthetics: The system should be simple, without any distracting or unnecessary content that could harm the effectiveness of the system. There will be more colorful tabs that will stand out from the rest of the page to indicate the buttons that they can press to take them to their desired section, so it is easy for both employees and customers to find the areas that they need.

User Experience: The goal is to create an all-encompassing system for PHP Agency’s employees and customers. While there will be a lot of information and a larger number of users, the goal is to simplify the interface as much as possible to allow users to efficiently gather the information and training they need.

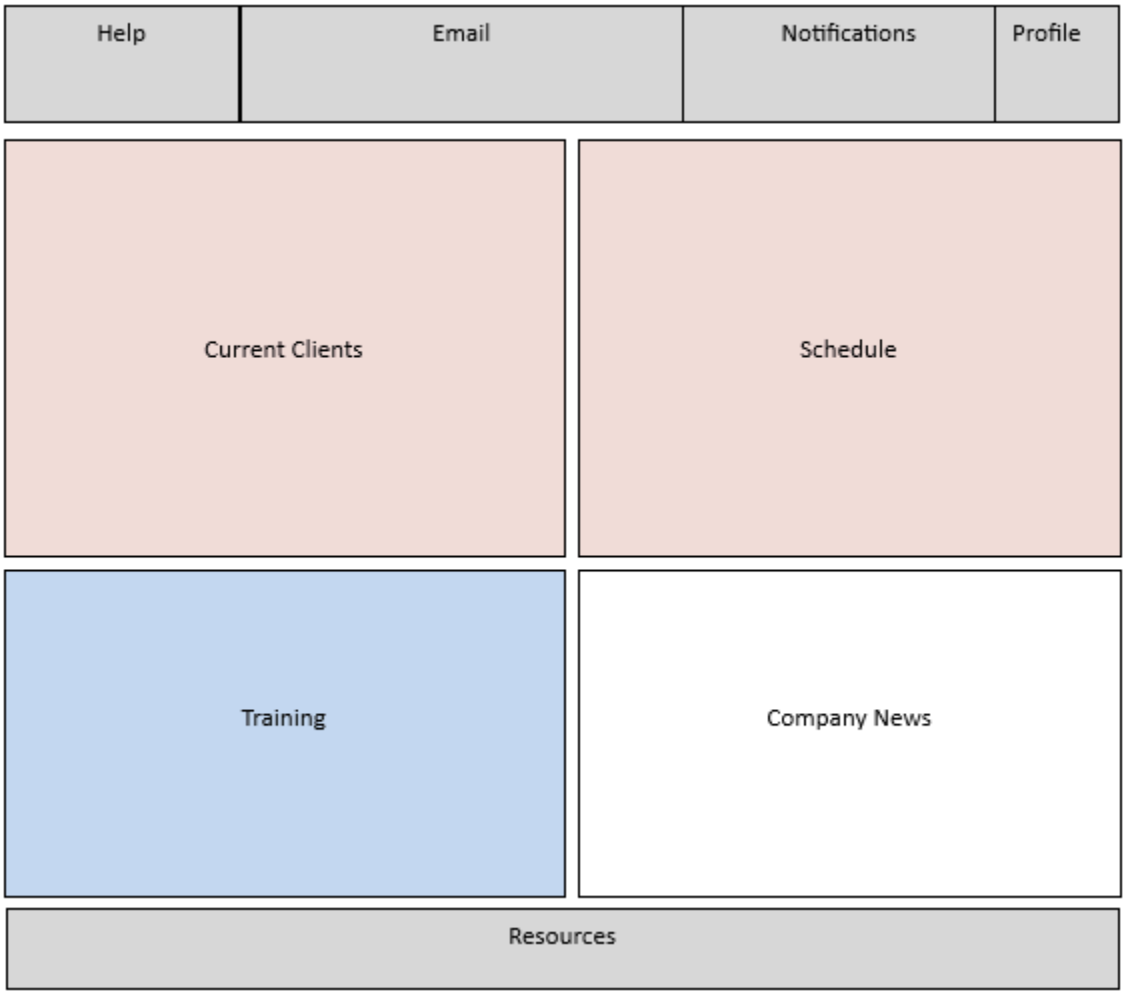
Consistency: Our system will be new to employees and customers because PHP Agency's current systems are far less compatible than the one that we are designing. As a result, it is necessary to make our system as simple as possible and maintain a consistent layout and design within the system, to ensure that the learning curve will be minimal.

Minimize User Effort: The goal is to make a low effort system that makes communication, scheduling, and training easy to find and access. The more straightforward we can make our system; the more PHP Agency's employees can focus on the jobs and hopefully our system provides assistance in that process without over complicating it.

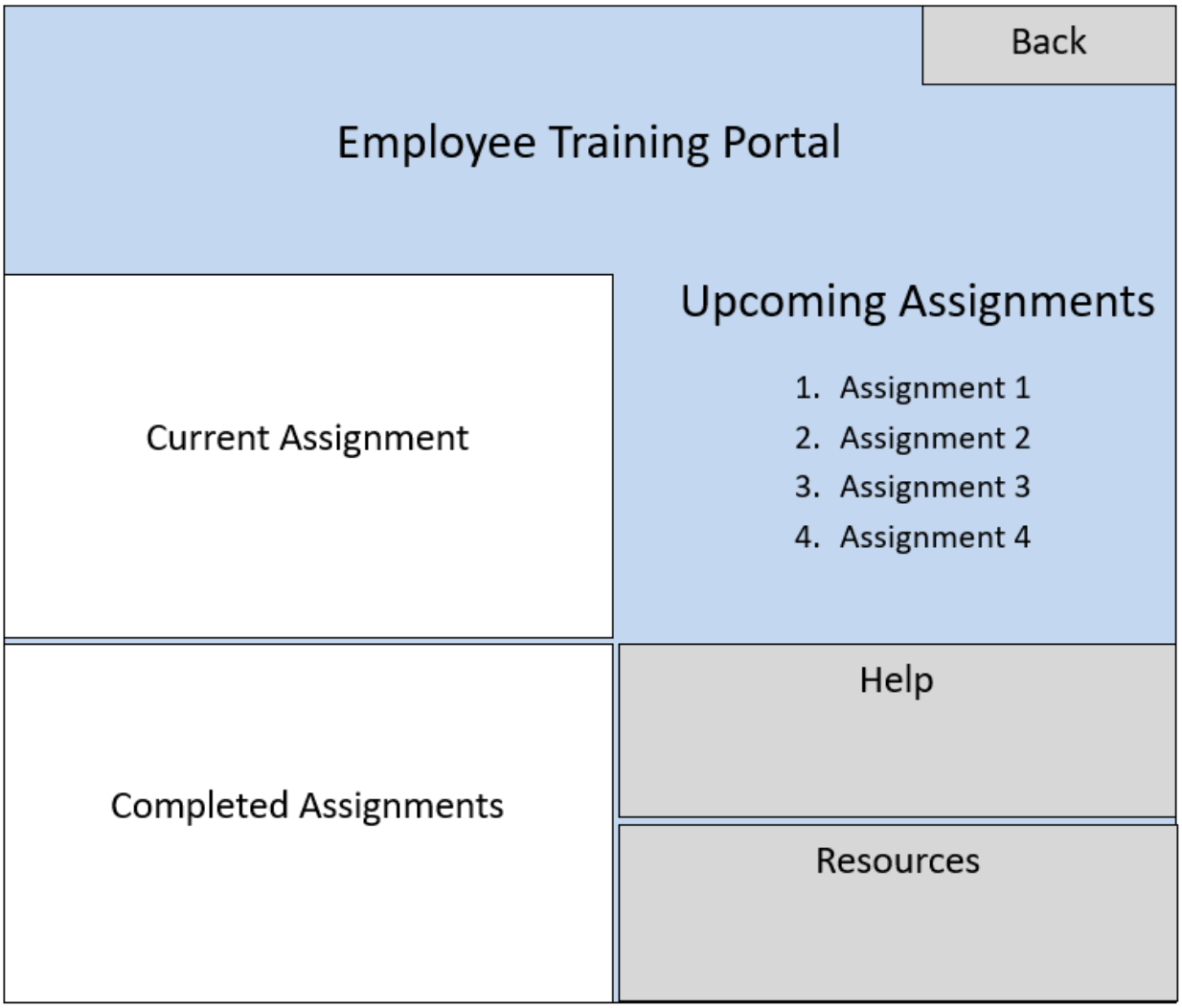
User Interface Design

The following designs are examples of how the interface will look from the perspective of the user. The icons on the interface represent buttons that can be pressed to take the user to their desired section. The examples below are for both employees and customers will use and examples of how the interface will look when a button is pressed including the customer profile page and employee training page.

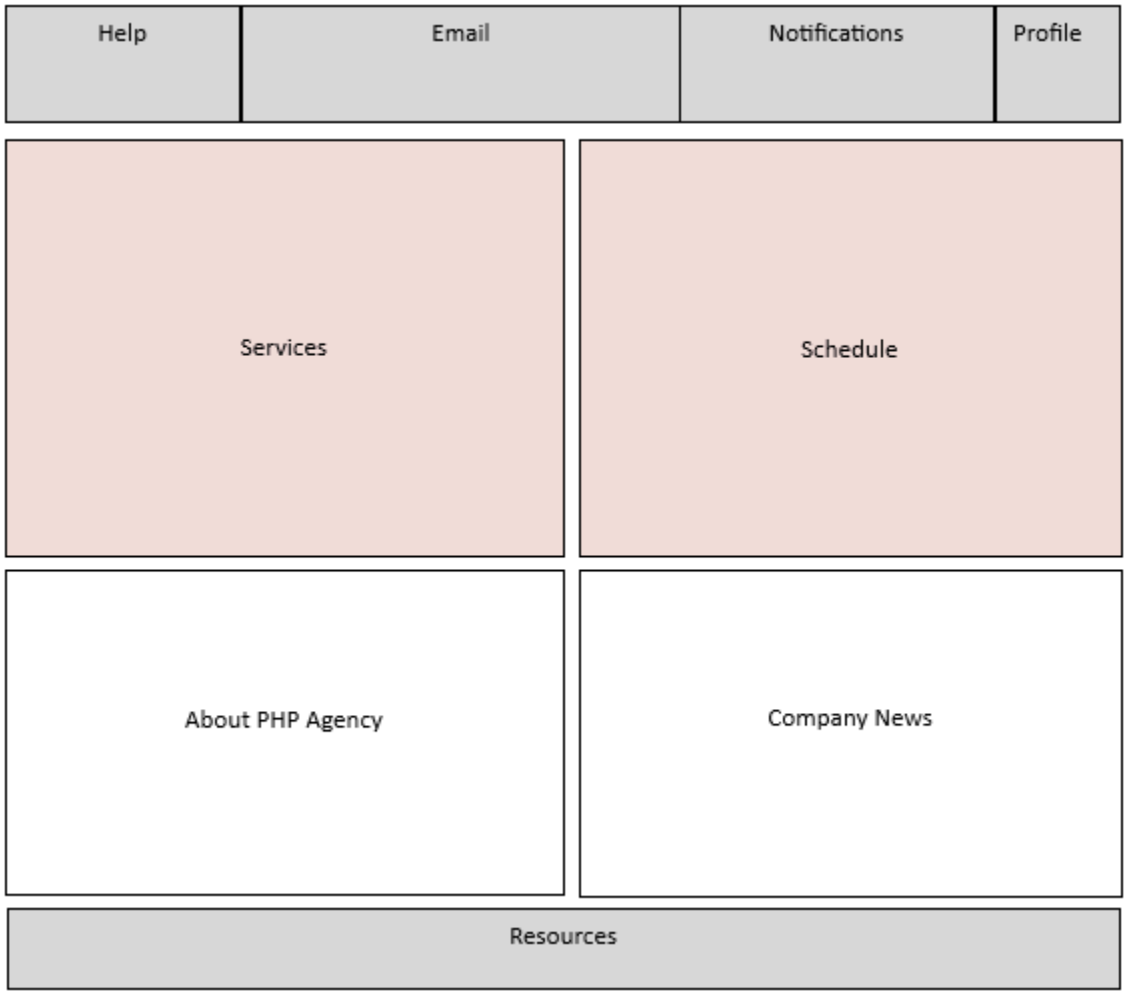
Employee User Interface Design



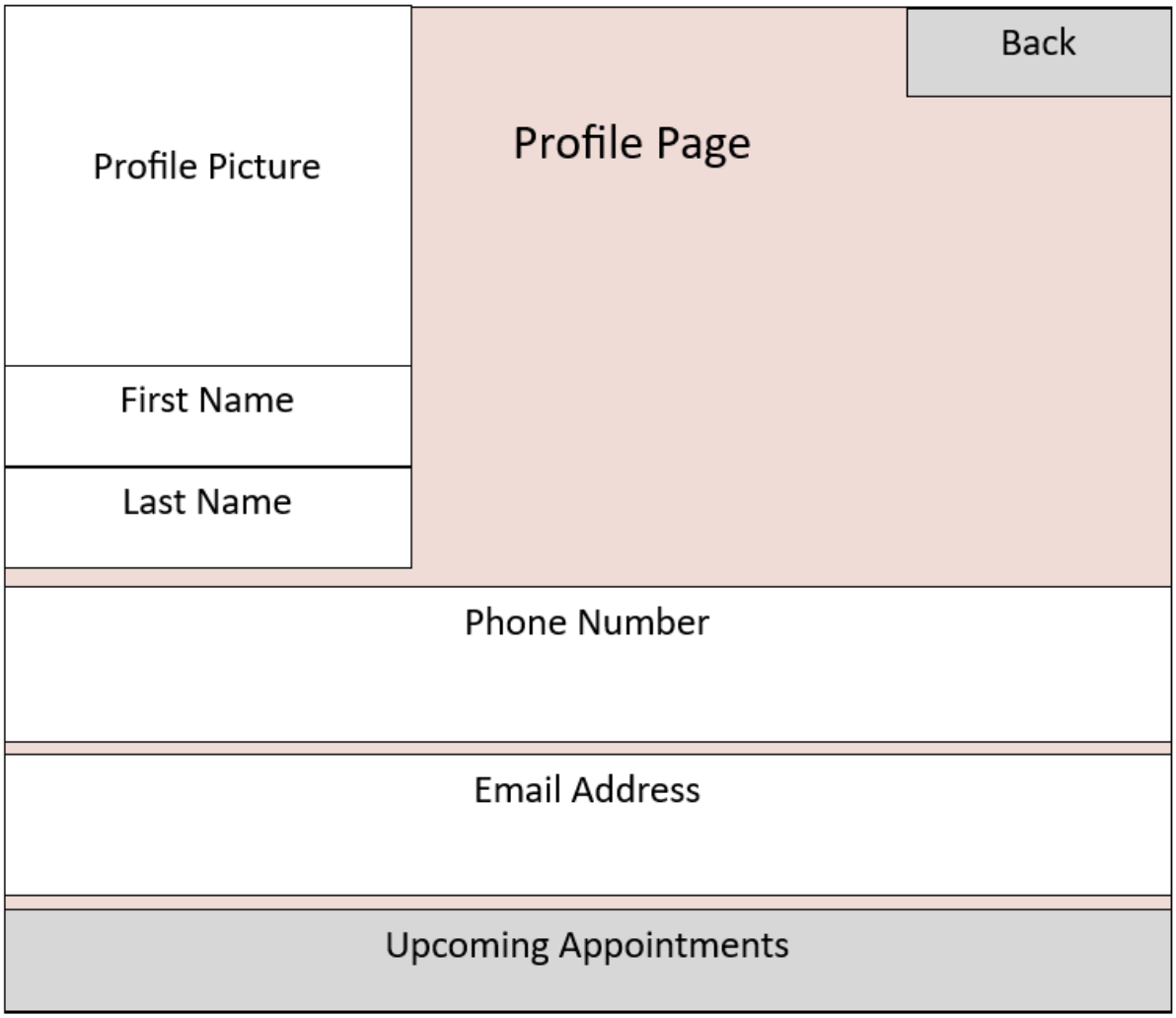
Employee Training Portal



Customer User Interface Design

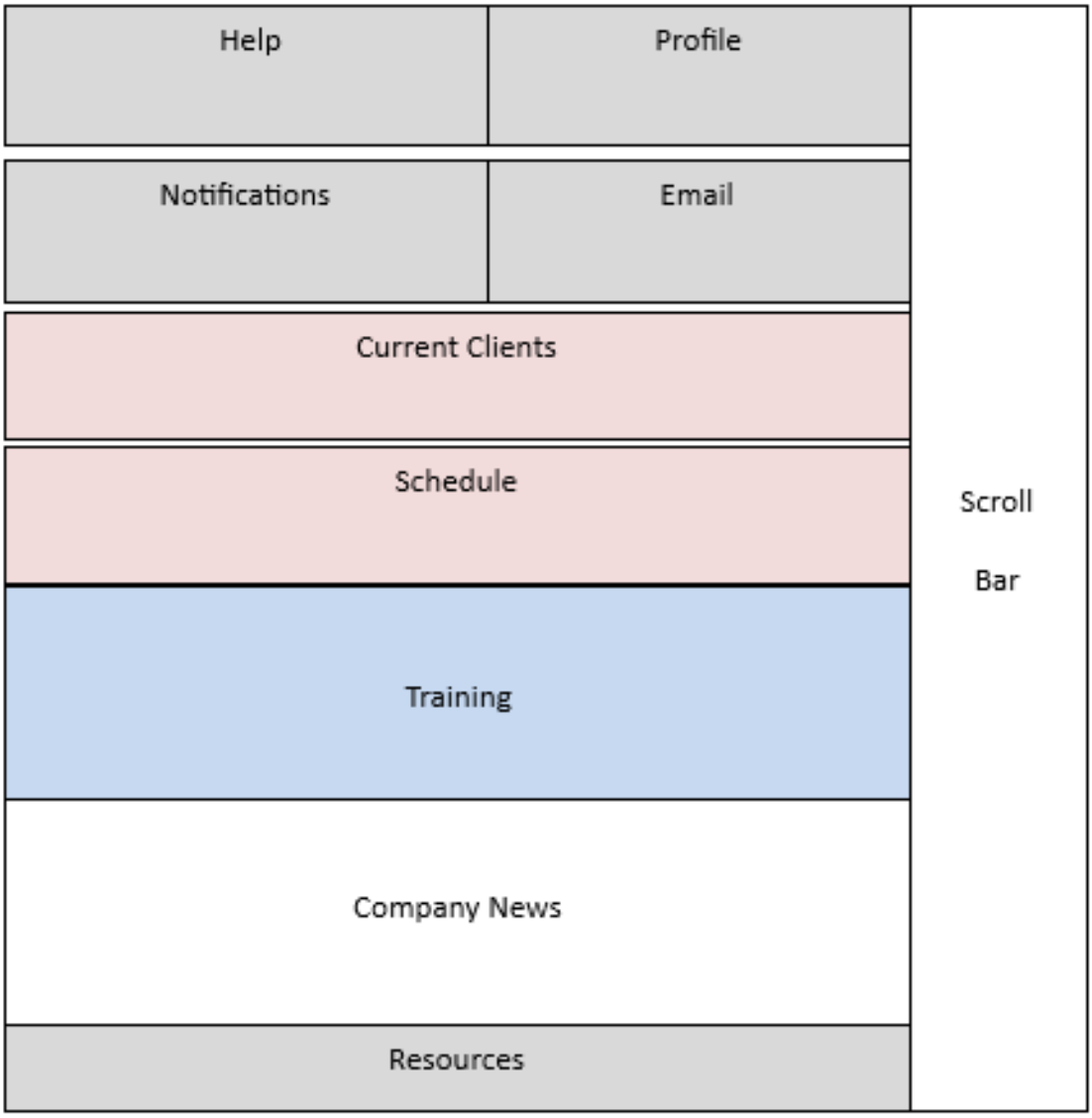


Customer Profile Page

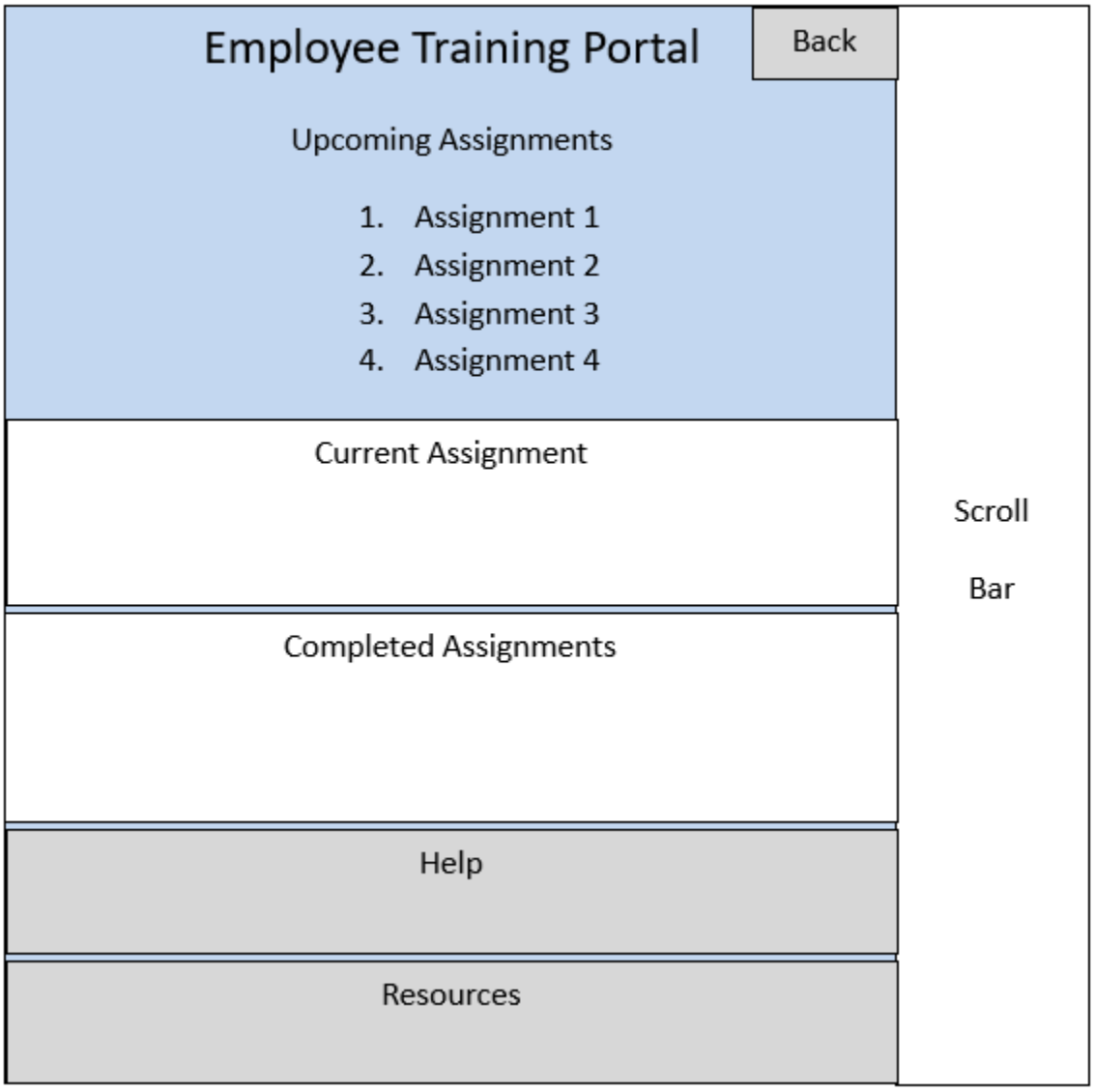


Handheld Device Design

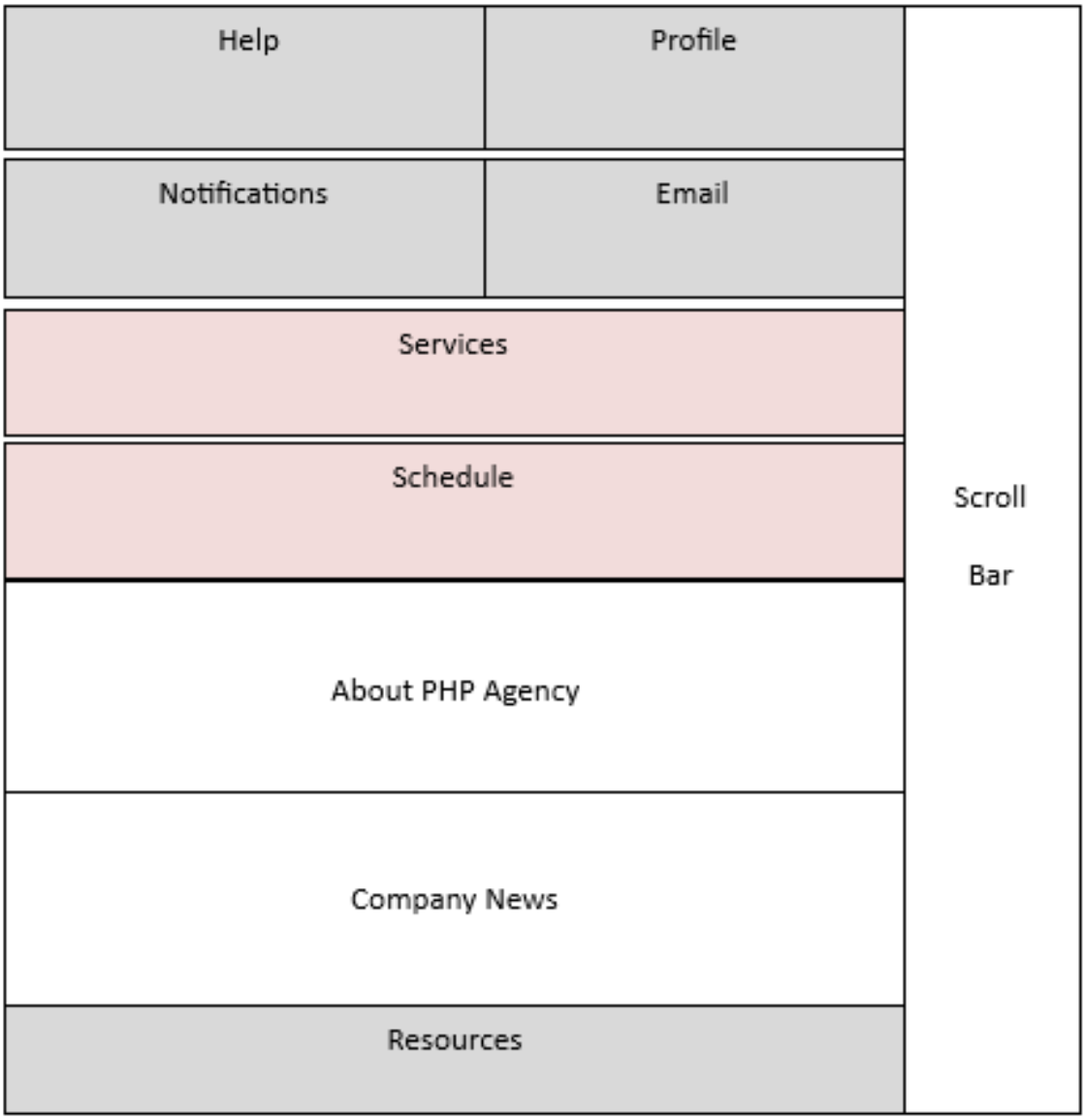
Handheld Device Design for Employees



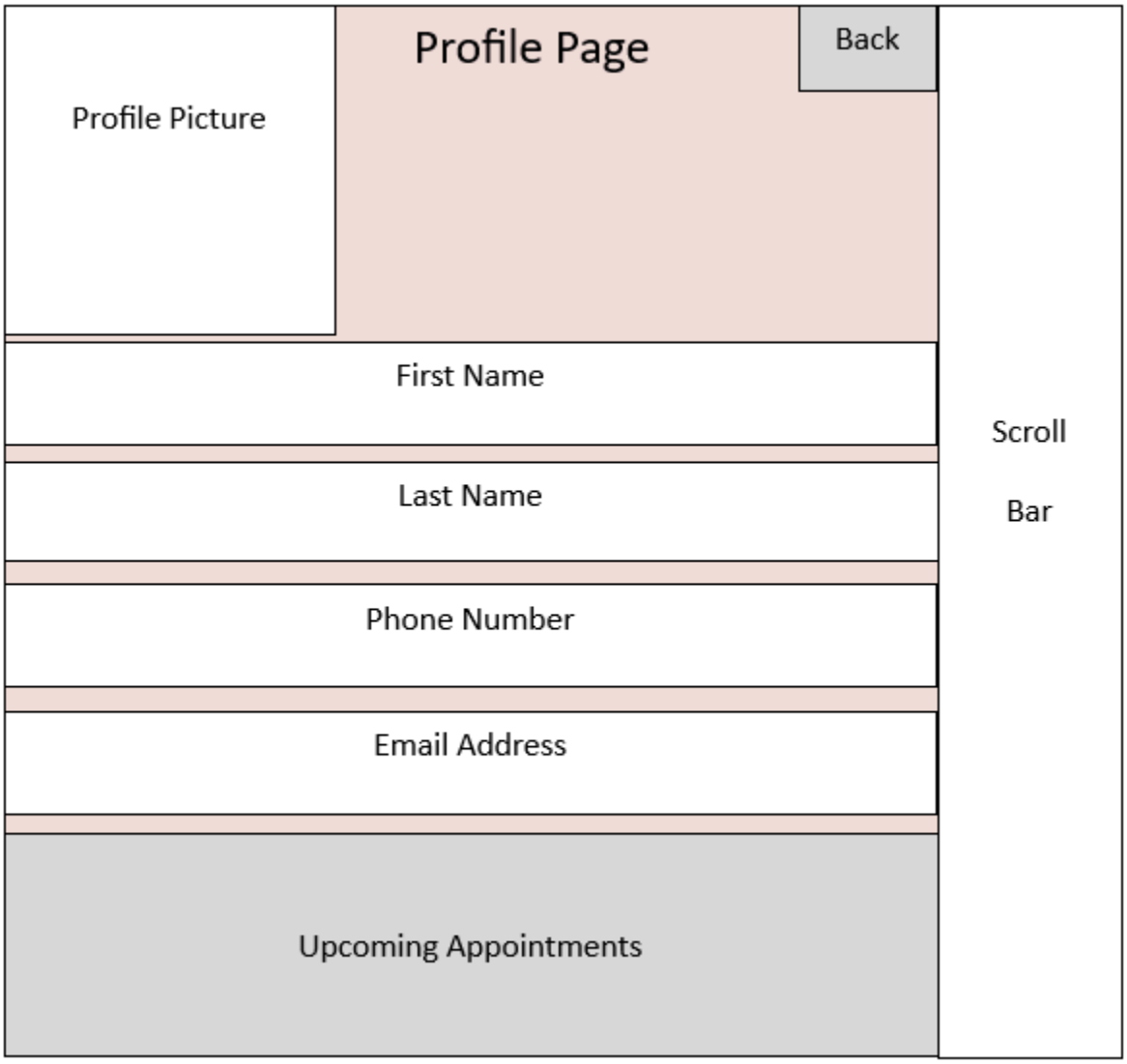
Employee Training Page on Handheld Devices



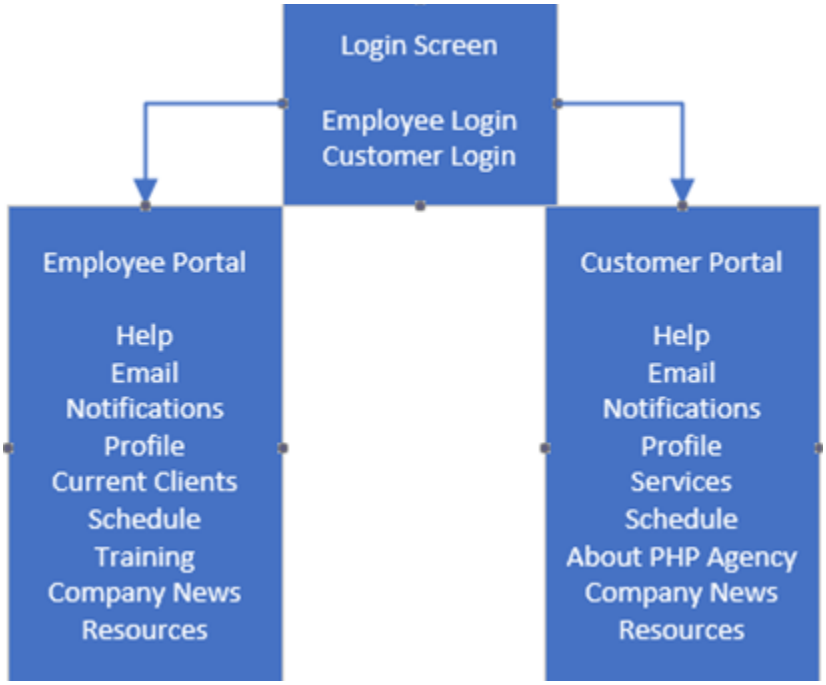
Handheld Device Design for Customers

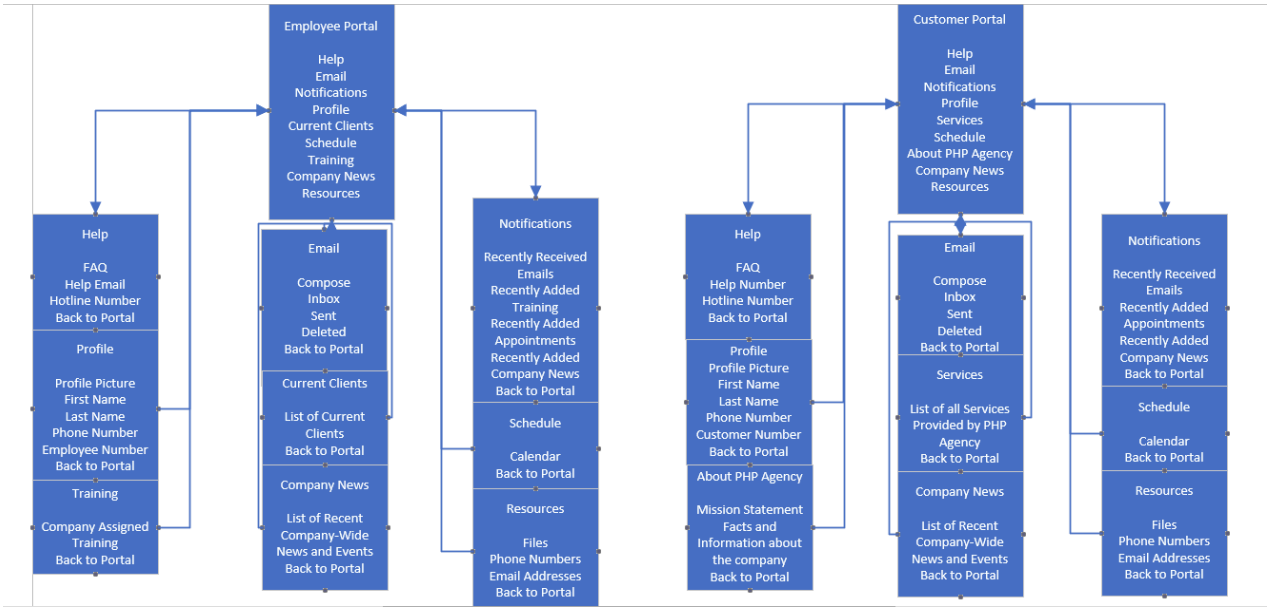


Customer Profile Page on Handheld Devices



Storyboard and Dialog Design





Dialog

Dialog Situation Examples:

Situation 1: New agent is making an account.
New Agent: Hello, I was recently hired by PHP Agency and need to get acclimated to the system.

Hiring Agent: Nice to meet you, we're happy to have you as a part of our team and glad to hear you are ready to get started. First, we need to create an account that will hold your important information and give you an employee ID.

New Agent: I just created an account, what are the next steps?

Hiring Agent: Great! After you have created your account, you can add a profile picture, and link your email address to the system to begin getting into contact with your clients and fellow agents.

New Agent: My email address is put in. What else can I do with this system to prepare me for my first week?

Hiring Agent: We have training modules and a company news tab that should teach you everything you need to know about PHP Agency and what we do, and if you need help, you can email a colleague or visit the resources tab to view files and get into contact with managers.

New Agent: Awesome! Thank you for your help.

Situation 2: New customer is making an account.
New Customer: Hello, I am in need of a new life insurance policy and was referred to the PHP Agency by a friend.

PHP Agent: Nice to meet you, we can certainly help you with that. The first step is to create an account in our system.

New Customer: My account is created, what should I do next?

PHP Agent: Next you can update your account by adding information to your profile. Include a profile picture, your name, email address, and phone number.

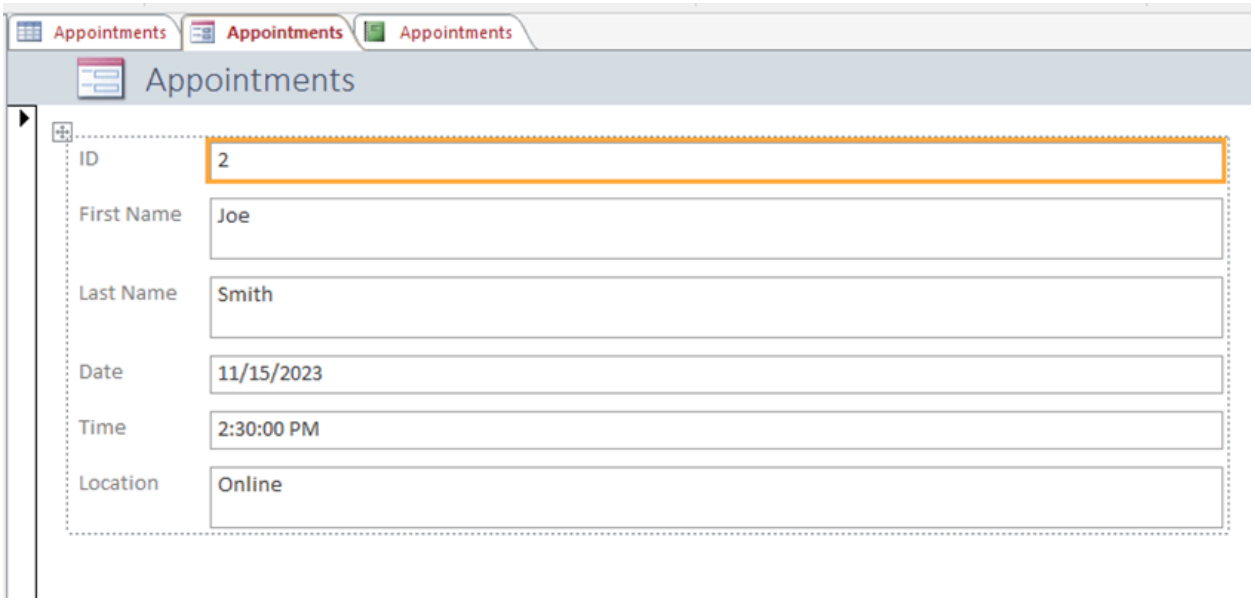
New Customer: I'm all set up, is there anything else I need to do?

PHP Agent: Yes, you can make an appointment to meet with an agent on the schedule tab and learn more about PHP Agency and the services we provide on the services, and about PHP Agency tabs.

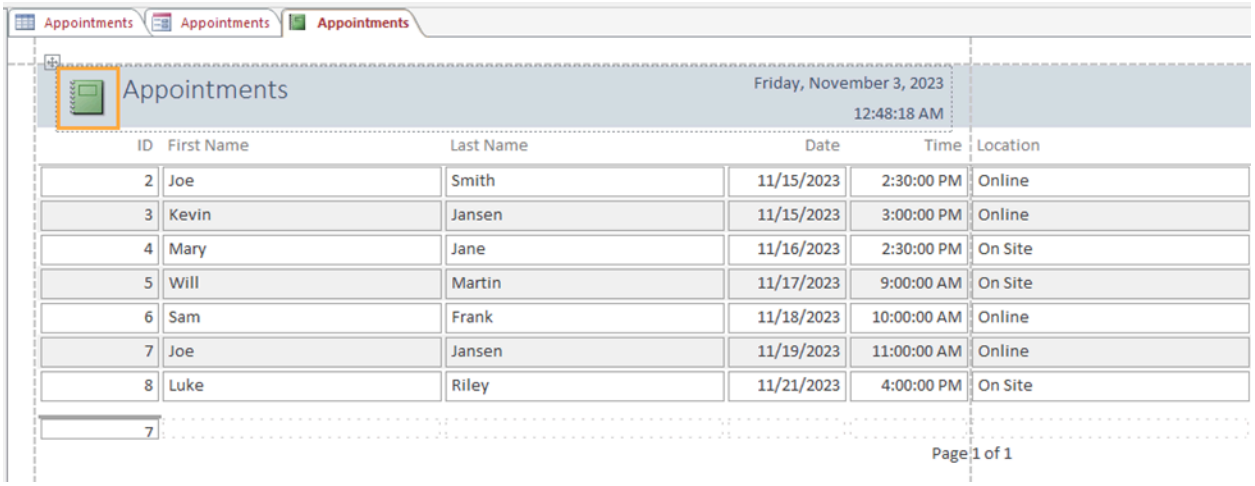
New Customer: Thank you very much!

Input and Output Designs

Sample Input Form
This input form allows PHP's agents to add appointments to their calendar and keep track of previously recorded appointments. The agents will input the first and last names of the customers, as well as the date, time, and location of the meeting.

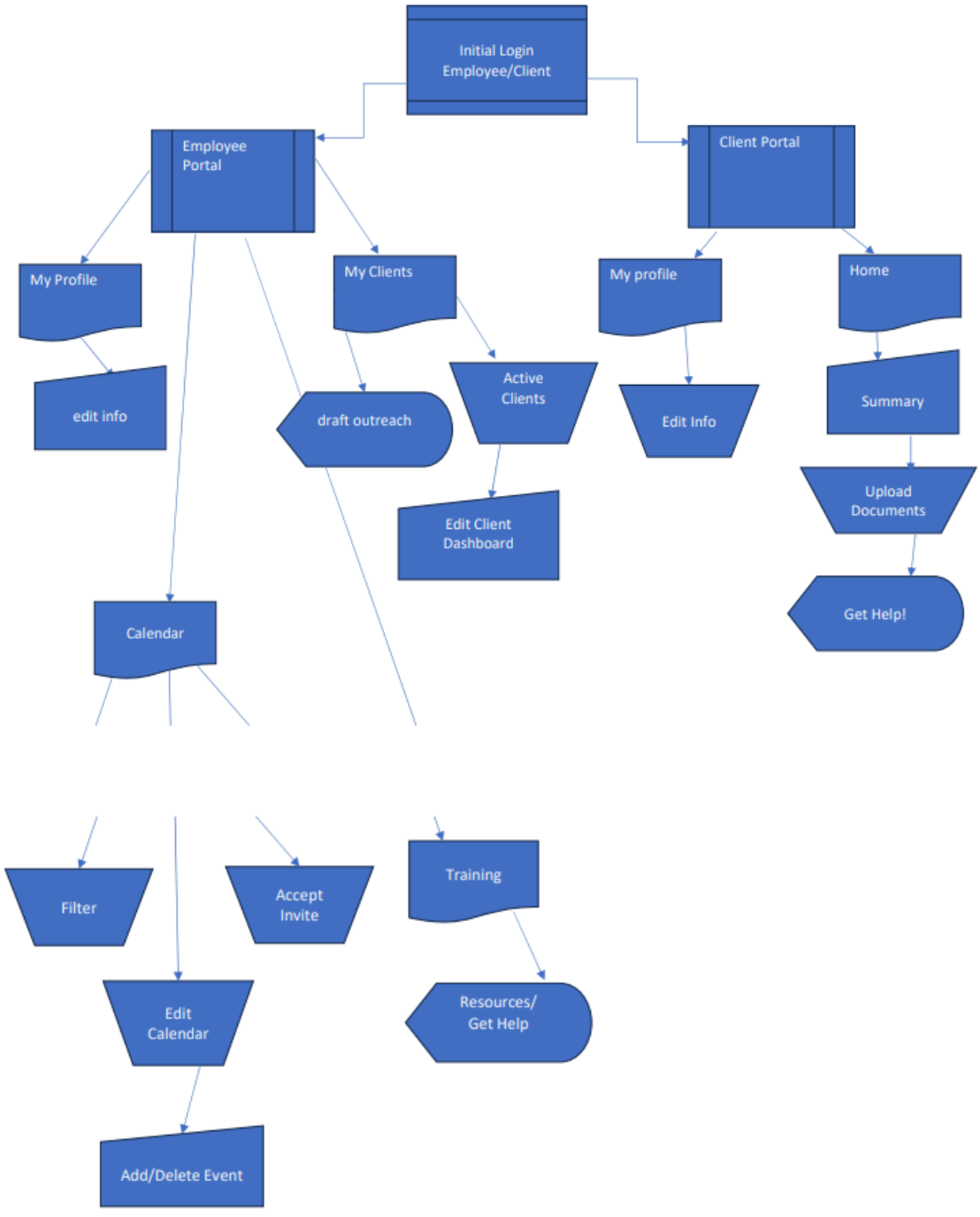


Sample Output Report
This output report can be viewed by PHP's agents as well as upper management to view currently scheduled appointments that have been created. This report consists of the customers' first and last names, and the date, time, and location of the meeting based on the data that was added to the input form.


Appointments
Appointments
Appointments
Appointments
Friday, November 3, 2023
12:48:18 AM
ID First Name Last Name Date Time Location
2 Joe Smith 11/15/2023 2:30:00 PM Online
3 Kevin Jansen 11/15/2023 3:00:00 PM Online
4 Mary Jane 11/16/2023 2:30:00 PM On Site
5 Will Martin 11/17/2023 9:00:00 AM On Site
6 Sam Frank 11/18/2023 10:00:00 AM Online
7 Joe Jansen 11/19/2023 11:00:00 AM Online
8 Luke Riley 11/21/2023 4:00:00 PM On Site
7
Page 1 of 1

Software Design

Structure Chart



Program Specification

The module outlines for each module are provided below, offering programmers the comprehensive details necessary for coding purposes. These outlines effectively depict the structure and requirements of the new system.

Program Specification 1.1 PHP Agency CRM System

Module: 1.1

Name: Initial Login: Employee/Client

Purpose: Facilitate log ins for both clients/employees

Programmer: Sheila Ramos

Due Date: November 3, 2023

Events:
Search box is selected.
Input Customer information: Client Name, Client DOB, Client Recruitment Status, Client Data

Provided By: Program 1.1

Output Name: Client Name, Client DOB, Client Recruitment Status, Client Data

Used By: Program 1.1

Pseudocode: (Single Patron module)

```
ENTER client_name, client_date_of_birth, client_recruitment_status, client_data
READ client_name, client_date_of_birth, client_recruitment_status, client_data
SAVE client_name, client_date_of_birth, client_recruitment_status, client_data
DISPLAY client_name, client_date_of_birth, client_recruitment_status, client_data
END
```

Program Specification 1.2 PHP Agency

Module: 1.2

Name: Employee Portal

Purpose: Tracking clientele, calendar events, and training resources

Programmer: Sheila Ramos

Due Date: November 3, 2023

Events:
Search box is selected.
Input New Client: Client name

Provided By: Program 1.2

Output Name: Client name, client calendar event, employee resources

Used By: Program 1.2

Pseudocode:
(Batch clientele tracking)
READ client_name, client_calendar_event, employee_resources
FOR all Client_name, client_calendar_event, employee_resources
    END FOR
END

Program Specification 1.3 PHP Agency

Module: 1.3

Name: Client Portal

Purpose: Uploading Documents and Information Resources

Programmer: Sheila Ramos

Due Date: November 3, 2023

Events:
Documents box is selected
Documents type is selected

Input name: Report type
Provided By: Program 1.3

Output Name: Documents
Used by: Program 1.3

Pseudocode:
(Documents module)
READ documents type
IF documents_type= ‘Uploading Documents’ THEN
    CREATE uploading_documents
ELSE IF documents_type= ‘Information Resources’ THEN
    CREATE information_resources
END IF
SAVE documents
DISPLAY documents
END

Pseudocode
(Single Patron module)
ENTER client_name, client_date_of_birth, client_recruitment_status, client_data
READ client_name, client_date_of_birth, client_recruitment_status, client_data
SAVE client_name, client_date_of_birth, client_recruitment_status, client_data
DISPLAY client_name, client_date_of_birth, client_recruitment_status, client_data
END
(Batch clientele tracking)
READ client_name, client_calendar_event, employee_resources

```
FOR all Client_name, client_calendar_event, employee_resources
        END FOR
END
(Documents module)
READ documents_type
IF documents_type= ‘Uploading Documents’ THEN
        CREATE uploading_documents
ELSE IF documents_type= ‘Information Resources’ THEN
        CREATE information_resources
END IF
SAVE documents
DISPLAY documents
END
```

Database Controls & Security Design

Input Controls

Data Validation:

- Customer Information: Validate data such as customer names, addresses, and contact details to ensure accuracy. Verify that all mandatory fields are filled out correctly and that formats, like email addresses and phone numbers, are valid.
- Policy Details: Validate policy data to ensure that it meets business rules, such as policy type, coverage limits, and premium amounts.
- Claim Data: Validate claim information to check for completeness and accuracy, helping to prevent fraudulent or incomplete claims.

Authentication and Authorization:

- Implement strong user authentication mechanisms for employees, agents, and customers interacting with the system. This may include username and password combinations or multi-factor authentication for added security.
- Role-based access control ensures that only authorized personnel have input access to specific areas of the database, limiting the potential for unauthorized data entry.
- Input Sanitization:
- Ensure that data received from online forms or external sources is sanitized to remove potentially harmful characters, scripts, or code. This prevents cross-site scripting and other injection attacks.
- Use input validation libraries to filter and cleanse data received from web forms and documents, reducing the risk of vulnerabilities in your system.

Encryption:

- Sensitive customer information, such as Social Security numbers and financial data, should be encrypted during input to the system. Use strong encryption algorithms to safeguard this data.

- Implement encryption for data transfer between your website and the database, especially during online transactions where sensitive information is exchanged.

Output Controls

Access Control:

- Control who can access and retrieve data within the system. Only authorized employees or agents should have access to customer records, policy data, and claims information.
- Implement strict access control mechanisms to ensure that employees only view data related to their responsibilities.

Data Masking:

- Implement data masking to hide sensitive information, such as Social Security numbers or full policy details, from employees or agents who don't have authorization to view this data. This helps protect customer privacy and ensures compliance with data protection regulations.

Logging and Auditing:

- Keep detailed logs of who accessed the database and what data they retrieved. Include timestamps and user identification to track any unauthorized access attempts.
- Regularly review audit logs to identify suspicious activities and potential security breaches.

Data Encryption:

- Encrypt customer data at rest within the database to protect it from unauthorized access. Employ encryption at the database level to ensure that even if a breach occurs, the data remains encrypted and unreadable.

PHP Software Input/Output Controls

Step 1: Accessing the Customer Portal
If you are a customer, access the PHP life insurance company's customer portal, which is typically available on the company's website.

Step 2: Registration and Login
If you are a first-time user, register for an account on the customer portal. Provide your personal details, policy information, and create login credentials.
Log in to your customer account using your registered username and password.

Step 3: Accessing Customer Services
After logging in, you will be presented with options to access various customer services. These may include:

- Learning about your insurance policies
- Making appointments with insurance agents
- Inputting information related to your case

Step 4: Learning About Your Insurance Policies
To learn about your insurance policies, click on “My Policies" or a similar option. Here, you can access detailed information about your life insurance policies, including coverage, premium amounts, and policy documents.

Step 5: Making Appointments
To make an appointment with an insurance agent or representative, select the "Appointments" or "Schedule an Appointment" option. This will typically open a calendar or booking interface.
Choose a convenient date and time for your appointment from the available slots. Some systems may allow you to specify the purpose or type of appointment.
Confirm your appointment, and the system will provide you with confirmation or appointment details. You may receive reminders via email or SMS, depending on the system's features.

Step 6: Inputting Case Information
If you need to provide information related to your case before your meeting with an employee, navigate to the "My Case" or "Submit Case Information" section. Here, you can input details about your inquiry or situation.
Complete the case information form, providing accurate and relevant details. This may include describing the nature of your case, any supporting documents, or specific questions you have.

Step 7: Review and Submission
Review the information you've entered to ensure its accurate and complete. Make any necessary edits or additions.
Once you are satisfied with the case information, submit it through the portal. The system may generate a confirmation of your submission.

Step 8: Meeting with an Employee
Attending your scheduled appointment with the insurance agent or employee on the chosen date and time. You can discuss your case and receive guidance based on the information you provided.

Step 9: Logging Out
After completing your tasks on the customer portal, be sure to log out of your account to secure your personal and policy information.

Security Controls

Access Control:

- Implement role-based access control (RBAC) to restrict access to the database based on job roles. Agents, underwriters, and administrators should have different levels of access, ensuring that sensitive information is only available to those who need it.

Authentication:

- Employ strong authentication methods, such as password policies that require complex and regularly updated passwords. Consider implementing multi-factor authentication (MFA) to further verify user identities.

Security Updates:

- Regularly update the PHP framework, database management system, and server software with security patches and updates to protect against known vulnerabilities.

Firewalls:

- Use of firewalls to safeguard the database server from unauthorized network access. Configure rules to only allow necessary traffic while blocking potential threats.
- Intrusion Detection Systems (IDS):
- Deploy IDS to monitor for unusual activities, potential security breaches, and intrusion attempts. Configure IDS to trigger alerts or actions when it detects suspicious behavior.

Data Encryption:

- Encrypt data at rest within the database using strong encryption algorithms to protect it from unauthorized access. Additionally, ensure that data is encrypted during transmission to and from the database.

Disaster and Recovery Plan

Regular Backups:

- Implement automated backup processes to regularly save customer policies, claims data, and other critical information. We will consider both incremental and full backups to minimize data loss in case of a disaster.

Redundancy:

- Set up redundancy for critical systems, such as databases, web servers, and load balancers, to ensure high availability. Implement failover mechanisms to minimize downtime during system failures.

Disaster Recovery Site:

- Establish a disaster recovery site with replicated data to quickly restore operations in case of a major disaster. This site should have synchronized data, ensuring minimal disruption to business operations.

Testing and Documentation:

- Regularly test the disaster recovery plan, simulating various disaster scenarios, and documenting the procedures. Ensure that employees are familiar with the plan and know how to execute it in case of an emergency. Update the plan as necessary to adapt to changing circumstances.

Incorporating these detailed controls and measures into the operations of your PHP insurance company agency will help ensure the security, integrity, and availability of your data, as well as regulatory compliance and protection against potential threats and disasters.

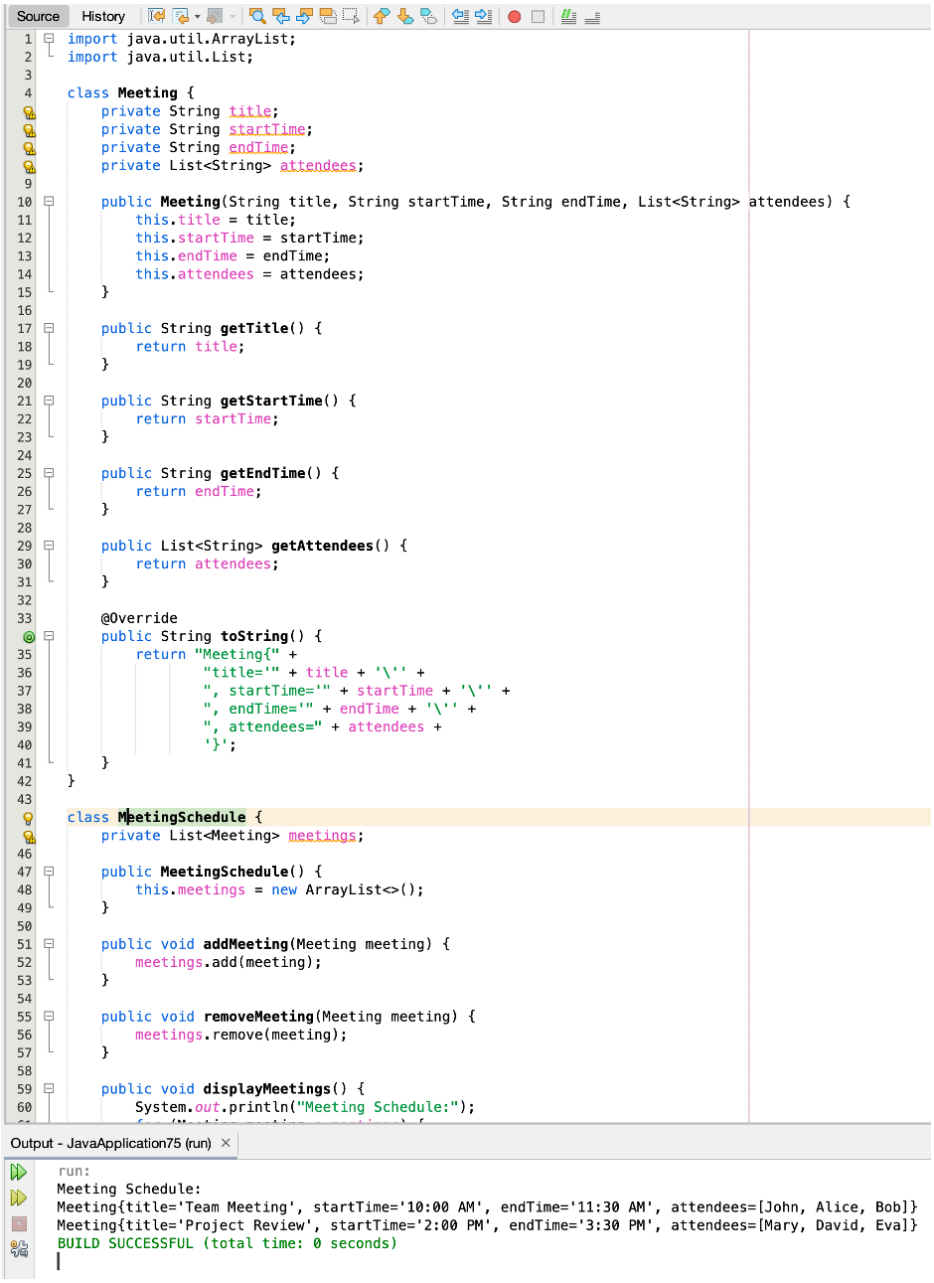
Phase 4 - Plans for Testing, Implementation & Training

Advanced System Analysis and Design

Testing plan

Unit Testing
Consider every possible part of this system and create some sample test plans for each. Groups should have between 3-5 actions per system module.

Successful Appointment/Meeting Schedule Unit Test



```
Source  History
1  import java.util.ArrayList;
2  import java.util.List;
3
4  class Meeting {
5      private String title;
6      private String startTime;
7      private String endTime;
8      private List<String> attendees;
9
10     public Meeting(String title, String startTime, String endTime, List<String> attendees) {
11         this.title = title;
12         this.startTime = startTime;
13         this.endTime = endTime;
14         this.attendees = attendees;
15     }
16
17     public String getTitle() {
18         return title;
19     }
20
21     public String getStartTime() {
22         return startTime;
23     }
24
25     public String getEndTime() {
26         return endTime;
27     }
28
29     public List<String> getAttendees() {
30         return attendees;
31     }
32
33     @Override
35     public String toString() {
36         return "Meeting{" +
37                 "title='" + title + '\'' +
38                 ", startTime='" + startTime + '\'' +
39                 ", endTime='" + endTime + '\'' +
40                 ", attendees=" + attendees +
41                 '}';
42     }
43 }
44
45 class MeetingSchedule {
46     private List<Meeting> meetings;
47
48     public MeetingSchedule() {
49         this.meetings = new ArrayList<>();
50     }
51
52     public void addMeeting(Meeting meeting) {
53         meetings.add(meeting);
54     }
55
56     public void removeMeeting(Meeting meeting) {
57         meetings.remove(meeting);
58     }
59
60     public void displayMeetings() {
61         System.out.println("Meeting Schedule:");
```

```
Output - JavaApplication75 (run)
run:
Meeting Schedule:
Meeting{title='Team Meeting', startTime='10:00 AM', endTime='11:30 AM', attendees=[John, Alice, Bob]}
Meeting{title='Project Review', startTime='2:00 PM', endTime='3:30 PM', attendees=[Mary, David, Eva]}
BUILD SUCCESSFUL (total time: 0 seconds)
```

Successful Email/Direct Messaging Unit Test

```java
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;

class User {
    private String username;

    public User(String username) {
        this.username = username;
    }

    public String getUsername() {
        return username;
    }

    @Override
    public String toString() {
        return "User{" +
                "username='" + username + '\'' +
                '}';
    }
}
```

```
Output - Message App1113 (run)
run:
Message sent from Alice to Bob: Hello, Bob!
Message sent from Bob to Alice: Hi, Alice!
Messages:
Message{sender=User{username='Alice'}, content='Hello, Bob!'}
Message{sender=User{username='Bob'}, content='Hi, Alice!'}
BUILD SUCCESSFUL (total time: 0 seconds)
```

Successful Contact List Unit Test

```java
import java.util.ArrayList;
import java.util.List;

class Contact {
    private String name;
    private String email;
    private String phoneNumber;
    private String address;

    public Contact(String name, String email, String phoneNumber, String address) {
        this.name = name;
        this.email = email;
        this.phoneNumber = phoneNumber;
        this.address = address;
    }
```

```
Output - contctlist1113 (run)
run:
Contacts:
Contact{name='John Doe', email='john@example.com', phoneNumber='123-456-7890', address='123 Main St'}
Contact{name='Jane Smith', email='jane@example.com', phoneNumber='987-654-3210', address='456 Oak St'}
BUILD SUCCESSFUL (total time: 0 seconds)
```

Integration testing

Other systems that our interface will interact with

For our system to successfully improve communication between PHP Agency's employees and customers, the user interface will need to interact with an email app that can be used from within our system. For this to work seamlessly, we will need to conduct an integration test between the email application and the email functions in the user interface. The integration testing will revolve around our system's ability to completely and accurately allow all users to work within the email component of the interface and be able to carry out any action that they would be able to do in the email application.

Testing plan

The email app will need to be tested using real life examples that simulate PHP Agency's employees daily use of their emails. This test will need to assure that all aspects of the email app's capabilities are fully operational within our system, including composition, receiving emails through the inbox, deleting emails, and saving drafts. This test will be successful if our system can transmit all data from each capability within the email application without errors. The integration testing plan will be carried out by simulating common actions that PHP Agency's employees will need to perform to do their jobs effectively. The test should allow us to find any error in the email application's compatibility with our system that could prevent certain actions from working. After the test is completed, we will run a full analysis of any errors that were found to resolve the problems with our interface. Finally, we will run the tests again to guarantee that all errors were fixed, and that full compatibility between the email application and our system has been achieved.

Actions for integration testing

We will test common actions that will be performed daily by PHP Agency's employees to find any errors in the email apps integration with our system. The test will result in a success if the actions have been performed and saved in both the email application and our system, with no conflicting data between the two systems.

Action 1: Sending Emails

Composing and sending emails is the most common action that will be performed by PHP Agency's employees, our test will consist of writing emails and sending them to one and to multiple recipients form within our user interface and validating whether those emails appear as sent in the email application. If both systems output the same data, then this portion of the integration test will be successful.

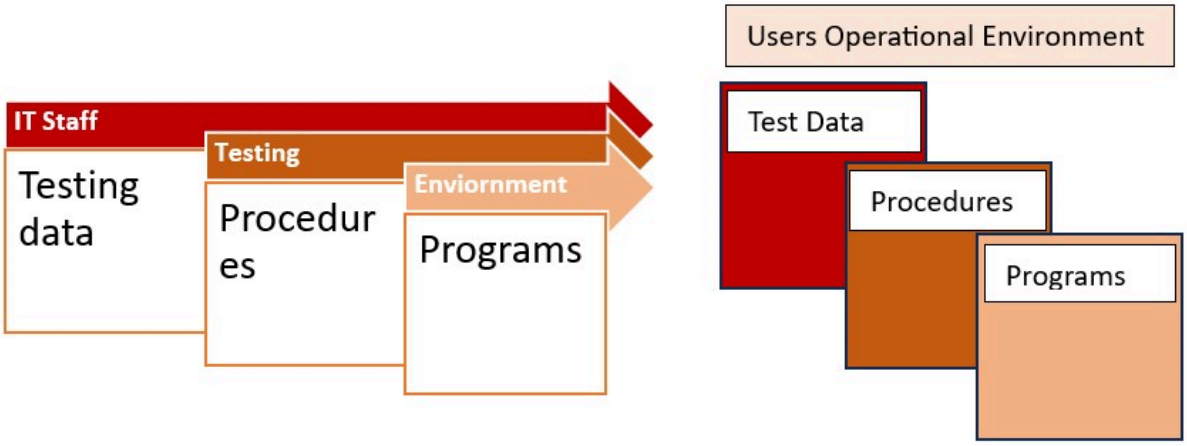
Action 2: Deleting Emails

Deleting emails is another important action that will be conducted every day by employees. In this portion of the integration test, we will simulate deleting received emails from within our system and check that those messages appear as deleted in the email application. For this action to be successful in our test, the deleted emails must appear as deleted in both systems and have no conflicting data.

Action 3: Saving Drafts

For the final portion of the integration test we will be reviewing the action of saving drafts in our system. Similarly, to the deleting emails portion of the integration test, we will test the functionality of saving drafts by executing this action from within our system and checking if the action was completed in the email application. The test will result in success if a saved draft in our user interface is saved in the email application and has the same information that was saved in the draft with no errors.

System testing



Introduction

Purpose

The purpose of this testing plan is to establish a structured approach to verify and validate the software, ensuring it meets the specified requirements and delivers a high-quality experience to end-users.

Scope

The scope of this testing plan encompasses various testing levels and types, including functional, usability, performance, security, and reliability testing. It covers the entire software, focusing on core functionalities, user interface design, system performance, data security, and overall system reliability.

Test Objectives

Functional Validation:

Verify that core features, such as task creation, collaboration tools, and customization options, function as expected.

Usability Assessment:

Evaluate the user interface for ease of use, intuitiveness, and accessibility.

Performance Evaluation:

Assess system response times, scalability, and resource usage under different scenarios.

Security Verification:

Ensure that authentication, authorization, and data protection mechanisms are robust.

Reliability and Stability:

Verify the software's stability under normal usage and its ability to recover from potential failures.

Test Environment

Hardware

Define the minimum system requirements for end-users to ensure compatibility.

Set up a dedicated test server with configurations closely resembling the production environment.

Software

Identify supported operating systems (Windows, macOS, Linux) and browsers (Chrome, Firefox, Safari, Edge).

Specify compatible database systems, such as MySQL and PostgreSQL.

Test Levels

Unit Testing

Develop and execute unit tests to verify the correctness and functionality of individual software components.

Integration Testing

Validate the interaction and communication between different modules to ensure seamless integration.

System Testing

Conduct end-to-end testing to verify the overall functionality, user interfaces, and system behavior.

User Acceptance Testing (UAT)

Involve end-users to validate whether the software meets their requirements and expectations.

Test Types

Functional Testing

Define and execute test cases for key functionalities, including task management, collaboration, and customization options.

Usability Testing

Conduct tests to evaluate the user interface, navigation, and overall user experience across different devices and browsers.

Performance Testing

Develop and execute performance test cases to evaluate system response times, scalability, and resource usage.

Security Testing

Perform security assessments, including authentication, authorization, and vulnerability testing.

Reliability Testing

Assess the software's stability under continuous usage and its ability to recover from potential failures.

Test Cases

Functional Test Cases

Develop detailed test cases for task creation, collaboration features, and customization options.

Usability Test Cases

Outline test cases for navigation, accessibility, and consistency across different devices and browsers.
Performance Test Cases
Specify test cases for load testing, stress testing, and performance under various conditions.
Security Test Cases
Develop test cases for authentication, authorization, and vulnerability assessments.
Reliability Test Cases
Define test cases to evaluate system stability under continuous usage and recovery from failures.

Test Schedule
Milestones
Define testing milestones based on the project timeline, including specific phases and deliverables.
Resource Allocation
Assign testing resources, including personnel and testing environments, to ensure efficient testing execution.

Test Deliverables
Develop detailed test plans for each testing phase.
Document test cases and test scripts for traceability and repeatability.
Generate comprehensive test reports for each testing phase, highlighting findings and recommendations.

Risks and Contingencies
Identify potential risks related to testing and the software, including technical, resource, and schedule risks.
Develop contingency plans to address identified risks promptly.

Approval
Seek approval from project stakeholders for the testing plan.
Obtain sign-off to proceed with testing activities according to the outlined schedule.

Implementation plan

Decide between parallel, direct, phased, or pilot changeover strategies.

We've concluded that using a pilot changeover implementation would be the best route to take for PHP considering the nature of the system we've built. We will implement it with directors first, managers second, and agents and customers third. This way, the people with more senior roles and a better understanding of the company from the top-down will be able to learn the app and teach those below them in the organization's structure. Another reason for a pilot changeover is that if any issues arise, they will be rectified and opinions/suggestions will be taken into account before most customers experience them, and we want customers to have a seamless experience to increase the chances of doing business with PHP. The sooner all employees and customers begin using our system, the sooner their workflow will be optimized. Since there will

be critically thought-out and focused training, the learning curve will be relatively fast due to the user-friendly design of our CRM. Our company's specialists will be available on-call to assist with any training for the new system as necessary. The complete changeover to our CRM is estimated to take place within a month of the initial implementation.

Training plan

Users

Introduction
Overview of CRM and its importance in recruitment and customer management, with an explanation of how the CRM system integrates with the recruiting process.

System Navigation
This part of the training session will consist of a CRM website interface tour that includes basic navigation skills, including accessing different modules and sections.

User Roles and Permissions
Understanding different user roles and their permissions while also ensuring users are aware of their access levels and limitations.

Client Data Entry
Users will learn how to add new client profiles to the CRM system and input essential information such as contact details, job history and relevant notes.

Recruiting Module
Users will explore the recruiting features within the CRM system where the steps for creating client listing, tracking applications, and managing client interactions.

Interactions and Notes
Users will record and track client interactions within the CRM system; this includes adding notes, comments, and follow-ups to maintain a comprehensive customer history.

Troubleshooting
Users will learn about common issues users may encounter and how to troubleshoot them. Will be notified of the contact points for support and additional resources.

Ongoing Training and Updates
Users will be aware of the importance of staying informed about system updates, with periodic refresher courses and additional training for advanced features.

Clients

Introduction to CRM system

Clients will get an overview of the system and its role in enhancing worker relationship management while also understanding the benefits of using the CRM system for effective collaboration.

Access and Login
Clients will receive guidance on accessing the CRM website and the steps for logging in and ensuring secure access.

Dashboard Overview
Clients will get a tour of the dashboard and its key features, and the training plan will also highlight areas related to worker relationships and interaction processes.

Worker Profiles
Clients will get an explanation of how to navigate and view the workers profile who is assigned to them and get information about the information available including skills, performance and communication history.

Interactions and Communication
Clients will learn how to track and review interactions with workers and also be able to add notes and comments about conversations for a comprehensive view.

Data Access and Permissions
Clients will understand client-specific data access and permissions while ensuring clients are aware of their level of visibility and interactions within the CRM system.

Feedback Mechanisms
Clients will utilize feedback tools within the CRM for effective communication which encourages clients to provide feedback on workers and the overall recruitment process.
Mobile Access (optional)
Clients will be able to receive instructions on accessing the CRM website through mobile devices to ensure they can stay connected and manage relationships on the go.

Troubleshooting Tips:
Clients will get an explanation of common issues clients may encounter and how to troubleshoot them and also provide contact points for support and assistance.

Ongoing Training and Updates
Clients will be aware of the importance of staying informed about system updates, with periodic refresher courses and additional training in advanced features.

<u>IT Staff</u>

System Overview
Techs will get an introduction to the CRM system's purpose in managing recruitment and relationships and how the CRM integrates into the IT infrastructure.

System Architecture
Techs will get a tour of the CRM website's architecture and technical components to get an understanding of data storage, security protocols and server infrastructure.

Security Protocols
Tech will get training on the security features and protocols of the CRM system which will emphasize the importance of data protection and user authentication.

User Access and Permissions
Techs will learn to manage user roles and permissions within the CRM, ensuring IT staff have the necessary access to maintain and troubleshoot the system.

Data Management
Tech will learn the best practices for data entry, validation and maintaining data integrity and the procedures for regular data backups and recovery.

Integration Points
Techs will get an understanding of how the CRM system integrates with other IT tools and systems with troubleshooting integration issues and ensuring seamless data flow.

Customization and Configuration
Techs will get training in configuring the CRM system to meet specific organizational needs to understand customization options for fields, workflows, and reports.

API usage
Techs will get an overview of available APIs for extending CRM functionality and examples of how APIs can be leveraged for automation and integration.
Monitoring and Maintenance
Techs will monitor system performance and address any issues promptly while conducting routine maintenance tasks to optimize CRM performance.

Troubleshooting and Debugging
Techs will develop skills for identifying and resolving common issues while utilizing debugging tools and logs for efficient issue resolution.

User Support and Training
Techs will provide support to end-users in case of technical issues and conduct periodic training sessions for users on system features and updates.

Security Audits
Techs will conduct regular security audits to identify vulnerabilities and implement measures to address any security concerns.

Documentation
Techs will maintain comprehensive documentation for the CRM system and create user manuals and technical documentation for trouble shootings.

Collaboration with Development Team
Techs will collaborate with developers for any customizations or enhancements and ensure alignment between IT and development teams for seamless integration.

Phase 5 - System Maintenance and Security Risks

Advanced System Analysis and Design

System Maintenance Plan

Now that we have designed the new system for PHP Agency's employees and customers, and created an implementation plan along with testing plans, we will develop a plan to maintain the integrity of the new system by focusing on each of the four types of system maintenance to ensure that all potential problems are planned for and addressed before the systems launch.

First, we will prepare to perform corrective maintenance on our system by locating and solving errors within our code, as well as update the system regularly to guarantee that each feature is functioning as it was intended to. These updates will include any fixes that were made in the effort to debug coding errors. It will be critical to receive constant user feedback to help alert the IT department of any problems so that relatively minor errors can be solved quickly.

In addition to our regular updates, that will be performed to fix any coding issues. It will be important to continue to update the system with new features to keep the system feeling fresh, prolong the value of the system with respect to PHP Agency's initial investment in the software, and avoid obsolescence. To achieve this, we will need to develop new features that improve the ease of use and allow PHP Agency's employees to do their jobs more efficiently. User feedback will be a key aspect of this part of our maintenance strategy. As employees and customers use the system, they will be able to recommend new features that they may find useful or even necessary.

In order to perform perfect maintenance, it is recommended that PHP Agency continues to invest in the hardware that their employees use to keep up with the additional software features that will be added throughout the years of the system's life. This type of maintenance is far less urgent than other sections within our greater maintenance plan. However, if the hardware isn't upgraded over the years, more serious issues could arise making this form of maintenance equally important for PHP Agency's total cost of ownership.

The most important section of the system maintenance plan is preventative maintenance. The decision to use in-house system maintenance or to outsource could have a major impact on the quality and timeliness of preventative maintenance, which will be further addressed in the recommendation section of the plan. Whichever strategy that ownership decides, having a knowledgeable IT team that can constantly analyze the system for security breaches as well as be able to preemptively stop any attempts of unauthorized access to the system. This is the most crucial section of our maintenance plan for keeping both employees' and customers' information safe.

Recommendation for In-House System Maintenance

It is recommended that PHP Agency develops in-house solutions for system maintenance rather than outsourcing. Keeping system maintenance in-house represents long-term upside for the system's growth as well as better customer service throughout the system's life.

Outsourcing can provide a cheaper solution for system maintenance. The need to hire and train a maintenance team including administrators, programmers, managers, and analysts will not be an issue because outsourcing to a professional company that specializes in software maintenance could provide PHP Agency with a team of professionals at a much smaller price than creating an in-house solution. However, issues with outsourcing can arise regarding customer service. Having an in-house maintenance staff would allow employees and customers to report issues and provide feedback directly to the internal team at PHP Agency. Having an in-house team would allow the company to respond to issues quickly and provide customer support much faster than an outsourced team that manages systems for multiple companies.

While short-term cost effectiveness is a positive for outsourcing, the potential for the most growth, the best customer service, and long-term profits would be possible with in-house system maintenance. While the need for hiring and training administrators, programmers, managers, and analysts would be costly, it would be beneficial for the company to invest its resources into hiring these positions in-house so that the company has complete control over who is in charge of the system. As a result, top-level management can receive direct reports from those hired team members and achieve a level of direct input over the system and those who manage it, that outsourcing would eliminate. Since this system has been planned as a long-term solution for PHP Agency, the pros of hiring an internal IT maintenance team outweigh the pros of outsourcing in regard to total cost of ownership and quality of system maintenance throughout the life of the system even if it would be more expensive to PHP Agency in the first immediate years after the launch.

Security Risks for the New System

Physical Security Risks

Unauthorized Access to Physical Hardware:

- Risk: Unauthenticated individuals gaining access to the dedicated test server or other hardware components, potentially compromising sensitive customer data.
- Mitigation: Implement biometric access controls and surveillance systems at the data center. Ensure the physical security measures comply with industry regulations.

Data Center Security:

- Risk: Inadequate security measures at the data center, risking the exposure of insurance-related data to unauthorized parties.
- Mitigation: Choose a data center with certifications specific to the insurance industry, such as those compliant with HIPAA or other relevant regulations.

Network Security Risks

Unauthorized Access to Network:

- Risk: Hackers gaining unauthorized access to the network, potentially compromising confidential insurance policy details.
- Mitigation: Employ intrusion detection and prevention systems, conduct regular penetration testing, and encrypt sensitive data in transit.

Integration Points Vulnerability:

- Risk: Weaknesses in the integration points between the CRM system and other insurance-related tools/systems, leading to potential data leaks.
- Mitigation: Ensure secure APIs and connections, following industry standards for insurance data exchange. Regularly audit and test integration points.

API Vulnerabilities:

- Risk: Exploitable vulnerabilities in the APIs, leading to unauthorized access or manipulation of insurance-related data.
- Mitigation: Implement strong API security measures, including proper authentication, authorization, and encryption. Regularly update API security protocols.

Application Security Risks

Authentication and Authorization Weakness:

- Risk: Weaknesses in user authentication and authorization mechanisms, risking unauthorized access to sensitive insurance information.
- Mitigation: Implement multi-factor authentication, strong password policies, and conduct regular security assessments. Ensure compliance with insurance industry security standards.

Customization Security:

- Risk: Insecure customization options introducing vulnerabilities in the CRM system, potentially compromising insurance policy data.
- Mitigation: Train IT staff in insurance industry security best practices for customizations. Conduct code reviews and adhere to insurance regulatory requirements for data protection.

File Security Risks

Data Integrity and Backup:

- Risk: Data integrity issues or lack of proper backup procedures leading to potential loss of critical insurance policy data.
- Mitigation: Establish robust data backup and recovery procedures, with a focus on insurance data. Regularly test data restoration to ensure quick recovery in case of incidents.

Unauthorized Access to Client Data:

- Risk: Unauthorized access to sensitive client data, including insurance policy details, leading to privacy violations.

- Mitigation: Implement strict access controls, encryption for client data, and monitor user activities. Comply with insurance industry regulations regarding customer data protection.

User Security Risks

Insufficient Training Leading to Human Errors:

- Risk: Users, including IT staff, making mistakes due to insufficient training, leading to security vulnerabilities and potential mishandling of insurance-related data.
- Mitigation: Provide industry-specific training programs, including insurance data protection practices. Conduct regular awareness sessions for all user roles.

Poorly Managed User Roles and Permissions:

- Risk: Mismanagement of user roles and permissions, allowing users more access than necessary to insurance policy details.
- Mitigation: Regularly review and update user roles and permissions, following the principle of least privilege. Align user access controls with insurance industry regulations.

Procedural Security Risks

Lack of Security Audits:

- Risk: Failure to conduct regular security audits, leaving vulnerabilities undiscovered and potential non-compliance with insurance industry standards.
- Mitigation: Establish a routine schedule for security audits, specifically addressing insurance data protection. Ensure audit processes align with industry regulations.

Inadequate Documentation:

- Risk: Insufficient documentation leading to challenges in troubleshooting and compliance verification with insurance industry security standards.
- Mitigation: Maintain comprehensive documentation for the CRM system, emphasizing compliance with insurance regulations. Create user manuals and technical documentation specific to insurance data protection.

By considering the insurance industry's unique security and regulatory requirements, the CRM system can be developed and implemented with a heightened focus on protecting sensitive insurance-related information.

System Backup Method

The IT team has been able to outline the pros and cons of the corrective performance system backup method. As the agency continually grows and their staff expands the system will have to continue to evolve and be able to withstand the amount of data that will be inputted into the database. Having the system perform those corrective actions that system itself will be able to detect issues and be able to debug any coding issues within it. Not only the system itself is able to detect any issues, but also having the customer feedback feature will allow customers to report

and submit any issues to the IT team. Aforesaid, the structure will facilitate the internal team's solution method when it comes to specific problems with the structure. This would be the most cost effective and cost friendly method to be implemented by the PHP Agency. In regard to the in-house system maintenance decision as explained earlier, the best decision that would work in favor of PHP agency would be the in-house resolution.

PHP's Total Cost of Ownership for the next 5 years

Cost Element	Description	Year 1 ($)	Year 2 ($)	Year 3 ($)	Year 4 ($)	Year 5 ($)	Total ($)
Software License Fees	Initial purchase or subscription fees	16,000	16,000	16,000	16,000	16,000	80,000
Implementation Costs	Consulting, customization, and integration	12,000	8,000	4,000	3,200	2,400	29,600
Training and Onboarding	Employee training and onboarding costs	6,400	3,200	1,600	800	800	12,800
Data Migration	Moving data from existing systems to the CRM	8,000	4,000	1,600	1,200	1,200	16,000
Support and Maintenance	Ongoing technical support, updates, and maintenance	4,800	4,800	4,800	4,800	4,800	24,000
Hardware/Infrastructure	Servers, networking equipment, or additional hardware	4,000	-	-	-	-	4,000
User Licenses	Additional licenses as the user base expands	4,000	5,600	8,000	9,600	12,000	39,200
Integration Costs	Cost of integrating the CRM with other business systems	6,400	3,200	2,400	1,600	1,600	15,200
Downtime Costs	Potential costs associated with system downtime	2,400	1,600	800	400	400	5,600

Upgrades/Enhancements	Additional features or upgrades requiring extra investment	4,000	3,200	2,400	2,400	2,400	14,400
Total Cost of Ownership		76,000	49,600	41,600	40,000	41,600	248,800

This table extends the analysis over a 5-year period, including additional costs such as hardware, ongoing user licenses, and potential downtime costs. We will consider these figures as a starting point and adjust based on their specific circumstances and the evolving needs of their CRM implementation.

References

PHP Agency. (2023). *Life Insurance Company*
www.phpagency.com